Precision Remaining in a Frozen Score: Conditional Gains Beyond Scalar Prediction-Powered Mean Inference

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Abstract

Can confidence intervals shrink further when labeled data, unlabeled data and a predictor are fixed? We examine population means using score-only control variates. A variance decomposition explains when nonlinear score information remains after scalar tuning and why affine or uninformative scores offer no first-order gain in this class. Across 165,000 paired draw sets, a cross-fitted cubic reduces mean width by 47.5% relative to PPI++ in a designed nonlinear Gaussian benchmark, with coverage compatible with 95%; the three archived-data benchmarks show no practical gain. Rare-event normal intervals severely undercover, and small-label fits can be unstable. Separately, an independently trained bounded-data construction has a finite-sample coverage guarantee and shortens empirical-Bernstein intervals by 28.7% versus label-only in a high-signal synthetic setting. It loses to the finite-sample baselines on the archived datasets. The answer is conditional on score information, fitting cost, unlabeled sample size and the required validity regime; no universal interval-length optimum is claimed.

1 Introduction

An expensive labeling process often leaves a small set of gold-standard outcomes, a larger set of unlabeled observations, and a prediction model that is already fixed. Prediction-powered inference uses the model's scores to improve estimation while correcting their bias with gold-standard labels. In this setting, collecting more labels, replacing the model, and enlarging the unlabeled sample may all be impossible. What precision can still be gained from the same inputs?

We study the population mean. This target has a direct scientific interpretation and makes the source of an improvement transparent: the score can explain variation in the outcome even when its calibration is nonlinear. A scalar correction can miss this information. Conversely, a highly flexible fitted correction can create apparent precision by underestimating its uncertainty. Any useful answer must assess coverage alongside width, and must compare against an efficiently tuned scalar correction rather than only an uncalibrated predictor.

Our answer is conditional. Within a deterministic score-control class, the optimal control is a shrunken conditional mean. A richer fixed basis improves on an unrestricted scalar control only through the additional conditional-mean variation it explains. There is no first-order gain when the conditional mean is affine in the score, and no score-based gain when the score carries no conditional-mean signal. These statements characterize estimator variance within a stated class; they are not a universal lower bound on confidence-interval length.

We evaluate this reasoning through paired repeated-sampling experiments with a frozen score. We distinguish normal-approximation intervals, whose justification is asymptotic, from a bounded-data construction with a finite-sample guarantee. The latter pays for honest nuisance fitting through a training split and for distribution-free concentration through range penalties. Thus an improvement in leading variance does not by itself establish a shorter valid finite-sample interval.

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The contribution is a reproducible examination of how much precision remains after scalar tuning, and where it disappears. It combines an explicit variance decomposition, controls for scalar calibration and fitting, empirical-population benchmarks with archived predictions, and a separate finite-sample comparison. The proposed transformations are classical control-variate operations on an existing score; no predictor is retrained and no new labels are acquired.

2 Related work and positioning

Prediction-powered inference corrects an imputed estimate with labeled evidence about prediction error, allowing a pretrained black-box predictor to support valid inference [Angelopoulos et al., 2023]. Its framework includes both normal-approximation procedures and constructions based on finite-sample concentration; calling an interval "PPI" alone therefore does not specify its guarantee. Our unit-control baseline is the usual Gaussian mean interval, while the empirical Bernstein comparisons separately address finite-sample validity.

PPI++ introduces power tuning and asymptotically normal estimators for a broad class of targets [Angelopoulos et al., 2024]. For means, power tuning learns a scalar control coefficient. Its discussion also explicitly connects prediction-powered estimators to augmented inverse-propensity estimation and the conditional-mean nuisance function, including the $1/(1 + n/N)$ shrinkage factor. We use this established connection rather than claim a new semiparametric principle. Our narrower question concerns what a fixed low-dimensional nonlinear transformation of an already frozen scalar score can add beyond scalar tuning, after paying for fitting and checking coverage.

Cross-prediction-powered inference uses labeled folds to train prediction models while evaluating their corrections on held-out data [Zrnic and Candès, 2024b]. Here the original prediction model remains fixed: cross-fitting applies only to the small inferential regression on its scalar output. We consequently make no claim of introducing cross-fitting or of evaluating the full cross-prediction method with newly trained predictors. The empirical Bernstein construction uses the known-range inequality of Maurer and Pontil [2009]; its guarantee comes from independent training and concentration, rather than from a Gaussian approximation. The study's contribution is the controlled comparison, explicit conditions, and reproducible evidence at this fixed-input scope.

Stratified PPI exploits conditional heterogeneity through a prespecified partition, fixed within-stratum sample counts, and masses treated as known [Fisch et al., 2024]. Retrospective iid draws instead have random stratum counts and finite-N mass uncertainty; importing its allocation benefit or treating estimated masses as fixed would change this comparison. We do not evaluate an iid poststratified/bin-indicator control, and claim no superiority over it or all nonlinear controls. Active statistical inference improves precision by choosing which labels to acquire [Zrnic and Candès, 2024a]; label identities are already fixed here, so that acquisition advantage is outside our setting.

3 Problem and inferential scope

Let $S = f(X)$ be a scalar score from a frozen predictor. We observe $L = \{(Y_i, S_i)\}_{i=1}^{n}$ and $U = \{\widetilde{S}_j\}_{j=1}^{N}$, drawn independently and identically within and between samples from a common population. The target is $\mu = \mathbb{E}[Y]$. Each procedure receives exactly the same L, U and frozen f in each repetition. A procedure may fit an inferential transformation of S using the available labels, but may not acquire labels, replace f, or augment U.

"Fixed" describes the comparison of procedures on a draw. Frequentist coverage concerns repetitions of the sampling experiment; it does not assign a sampling probability to one fully conditioned-on data set. We do not study confidence conditional on every observed label and score, adaptive label acquisition, distribution shift, or uncertainty in a newly trained prediction model.

The empirical-data experiments use each archived record list as a discrete empirical population. Labeled and unlabeled indices are sampled independently with replacement, and the mean of all archived labels is the benchmark truth. This gives a known target and the same sampling model as the synthetic experiments. It does not establish inference for a broader scientific population. Repeated indices are permitted by that sampling law. The evaluation harness uses all archived labels to define truth; estimators receive only their selected labeled outcomes.

4 How much improvement is available?

4.1 A score-control variance identity

For deterministic h, consider

$$\widehat{\mu}_h = \frac{1}{n}\sum_{i=1}^{n}\{Y_i - h(S_i)\} + \frac{1}{N}\sum_{j=1}^{N} h(\widetilde{S}_j). \tag{1}$$

Under common-population sampling and finite second moments, this estimator is unbiased and has variance

$$V(h) = \frac{\mathrm{Var}(Y - h(S))}{n} + \frac{\mathrm{Var}(h(S))}{N}. \tag{2}$$

Let $r = n/N$ and $m(S) = \mathbb{E}[Y \mid S]$. Constants in h cancel in (1). Completing the square yields

$$\begin{aligned} V(h) = \frac{\mathrm{Var}(Y)}{n} - \frac{\mathrm{Var}(m(S))}{n(1+r)} \\ + \frac{1+r}{n}\,\mathrm{Var}\left(h(S) - \frac{m(S)}{1+r}\right). \end{aligned} \tag{3}$$

Thus $h^*(S) = m(S)/(1+r)$, up to a constant, minimizes variance in this class, giving

$$V^* = \frac{\mathbb{E}[\mathrm{Var}(Y \mid S)]}{n} + \frac{\mathrm{Var}(m(S))}{n+N}. \tag{4}$$

Even perfect prediction leaves finite unlabeled-sample uncertainty. Transforming one score repeatedly does not create new independent observations.

For scalar $h(S) = aS$, the optimum is

$$a^* = \frac{\mathrm{Cov}(Y, S)}{(1+r)\,\mathrm{Var}(S)} \tag{5}$$

when $\mathrm{Var}(S) > 0$. If the score is constant, it cannot help. Let $\Pi_1 m$ and $\Pi_p m$ be the centered L^2 projections of m onto a scalar score and a nested, fixed p-dimensional basis. The decrease in optimal variance is

$$V_1^* - V_p^* = \frac{\mathrm{Var}(\Pi_p m) - \mathrm{Var}(\Pi_1 m)}{n(1+r)}. \tag{6}$$

An affine conditional mean leaves no additional projection signal. A nonlinear conditional mean can yield a strict gain, but only if the chosen basis represents its additional variation. These oracle comparisons do not pay for estimating coefficients and do not imply that fitted intervals improve for every n.

4.2 Fitted controls and normal-approximation intervals

Our richer control uses three prespecified Legendre coordinates on $[0, 1]$, excluding the intercept. Two fixed equal folds alternate between fitting centered least squares on one half of the labeled observations and evaluating residuals on the other half. Each fitted coefficient vector is multiplied by $1/(1 + n/N)$. The estimator averages the held-out corrections and uses the corresponding average control on all N unlabeled scores.

The standard error combines the empirical variance of out-of-fold residuals divided by n and the variance of the averaged unlabeled control divided by N. We use two-sided Gaussian intervals. With fixed basis dimension, nonsingular population basis covariance, consistent coefficients, finite second moments, positive limiting oracle variance, and $n, N \to \infty$ at positive finite ratio, the product of coefficient error with the held-out labeled/unlabeled basis-mean difference is $o_p(n^{-1/2})$: coefficient consistency gives $o_p(1)$ error, while the mean difference is $O_p(n^{-1/2})$. The interval then has the oracle fixed-basis normal limit. Cross-fitting protects held-out residual evaluation from direct training reuse; it does not give exact finite-sample coverage.

We compare label-only normal intervals, unit-control PPI, the pinned PPI++ scalar implementation, an unrestricted scalar, and cross-fitted linear and cubic controls. The unrestricted scalar separates

nonlinear gains from restrictions imposed by clipping the scalar tuning parameter to $[0, 1]$. Cross-fitted linear controls help separate nonlinear representation from the cost of splitting and fitting. A cubic fitted and evaluated on the same labels and prespecified degree-12 controls diagnose overfitting. No basis dimension is selected from the reported coverage or width results.

4.3 A finite-sample bounded-data construction

Suppose $Y, S \in [0, 1]$. For $k \geq 2$ independent identically distributed observations Z in a known interval of length R, with unbiased sample variance v_Z, define

$$b(Z, R, \delta) = \sqrt{\frac{2v_Z \log(4/\delta)}{k}} + \frac{7R \log(4/\delta)}{3(k-1)}. \tag{7}$$

Applying the empirical Bernstein inequality to Z and its reflection yields a two-sided interval with failure probability at most δ. The additive range term matters even if observed residual variance is small.

Reserve $\max(10, \lfloor n/4 \rfloor)$ labeled observations for fitting; let $m \geq 2$ be the remaining count and require $N \geq 2$. Conditional on training, choose a fixed control $h \in [0, c]$. Residuals lie in $[-c, 1]$, with known length $1 + c$. A valid radius for (1), now using m evaluation labels, is

$$b(Y - h(S), 1 + c, \alpha/2) + b(h(\widetilde{S}), c, \alpha/2). \tag{8}$$

A conditional union bound and then averaging over training give coverage at least $1 - \alpha$. Intersecting the interval with $[0, 1]$ preserves coverage. No correctness of the fitted regression is needed.

We use a clipped scalar and a clipped cubic regression trained only on the reserved labels, with shrinkage $1/(1 + m/N)$. All training labels count toward the original n-label budget. The label-only empirical Bernstein baseline uses all n labels. A unit-control baseline also uses all n labels because it requires no fitting; a common-split version exposes the split's cost. Binary outcomes additionally permit exact label-only Clopper–Pearson intervals. These comparisons address finite-sample validity directly; they should not be pooled with asymptotically justified Wald intervals as if all guarantees were interchangeable.

5 Experimental design

All methods are paired on identical draws. The main grid uses $n \in \{100, 500, 2000\}$, $N = 10n$, and 5,000 independent repetitions per setting. Seven synthetic regimes cover linear and nonlinear calibration, no signal, heavy-tailed noise, nonlinear Bernoulli responses and rare events. Three archived binary-outcome datasets supply frozen scores for galaxy morphology, forest loss and protein disorder. A separately specified high-signal bounded quadratic experiment uses $n \in \{2000, 10000\}$ and 3,000 repetitions to study whether finite-sample range penalties can be overcome. Generating mechanisms, seeds and sampling details are in Appendix C and the supplied code. The complete study contains 165,000 draw sets over 35 scenario/sample-size cells; all 1,935,000 method records are retained.

Coverage is the fraction of intervals containing the known mean; its uncertainty is a 95% Wilson Monte Carlo interval. Average widths have Monte Carlo standard errors. A relative width is the ratio of paired mean widths, not the average of within-repetition ratios. Its standard error uses the paired delta method. A narrow interval is not judged preferable when it undercovers. Coverage near 95% in a finite simulation is evidence about the tested setting, not a proof of validity. Exploratory one-sided exact-binomial undercoverage tests use Holm adjustment across all cells separately for each method. The family is the completed study grid, not a distribution-uniform test.

6 Results

Figures 1 and 2 pair coverage with width; Appendix E reports sample-size trends and diagnostics, and Appendix G retains every completed method/setting cell.

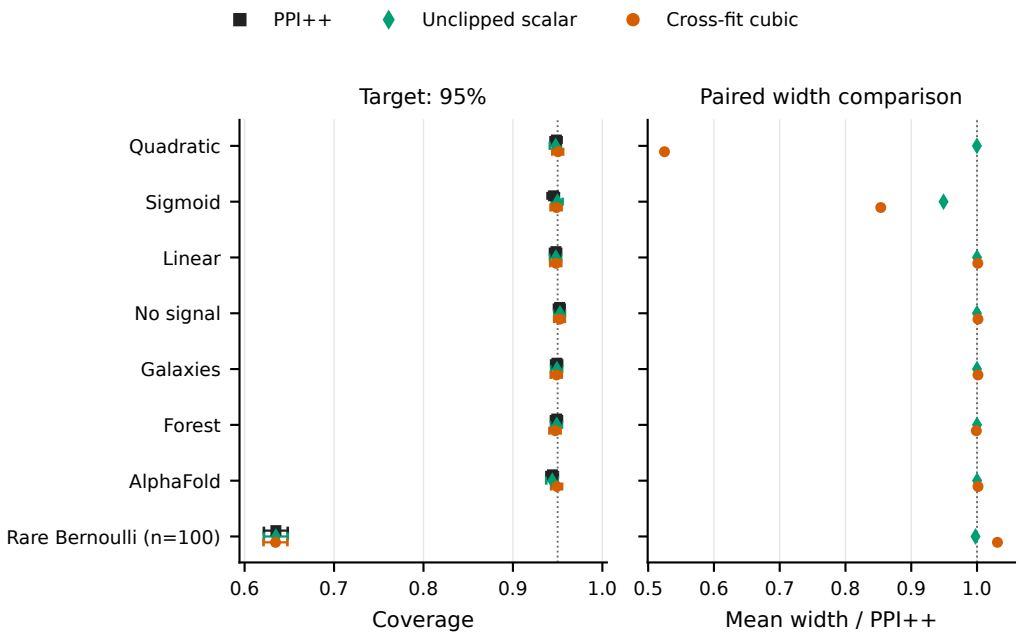


Figure 1: Nonlinear gains, no-gain controls, archive results and a small-sample failure. Scalar PPI++ (`ppi-python 0.2.3`), unrestricted scalar and cross-fitted cubic use identical inputs. All rows have $n = 2000$, $N = 20000$, except rare Bernoulli ($n = 100$, $N = 1000$); $R = 5000$ repetitions per row. Left: coverage with 95% Wilson Monte Carlo intervals. Right: ratios of paired mean widths to PPI++, with 95% delta-method Monte Carlo intervals. Rare-event short intervals fail. The archive rows use independent with-replacement draws and target empirical archive means, with no broader population claim. These Wald intervals have asymptotic justification, not finite-sample guarantees.

Nonlinear signal can leave substantial precision after scalar tuning. For quadratic Gaussian regression at $n = 2000$, $N = 20000$, the cubic/PPI++ ratio of mean widths is 0.5248 (paired 95% Monte Carlo interval [0.5246, 0.5250]), a 47.5% reduction. Cubic coverage is 0.9504 (Wilson interval [0.9440, 0.9561]); PPI++ coverage is 0.9486 ([0.9421, 0.9544]). Both are compatible with the nominal target, while neither has an exact finite-sample guarantee. Against the unrestricted scalar, the ratio is 0.5249 ([0.5247, 0.5251]), so clipping is not the explanation. The predicted leading-order oracle ratio is 0.524, closely matching the fitted result. Linear and no-signal cases instead give cubic/PPI++ ratios 1.0012 and 1.0015, respectively, consistent with a small fitting cost and no first-order nonlinear benefit. These controls matter as much as the designed positive case.

Finite unlabeled information limits the gain. A documented follow-up after the initial real/bounded results repeats linear, quadratic and no-signal regimes with $n = N = 2000$, using 3,000 repetitions. The quadratic ratio becomes 0.7754 ([0.7752, 0.7756]), close to the analytical prediction 0.775, with cubic coverage 0.9560 ([0.9481, 0.9628]). The reduction from the $N = 10n$ gain agrees with the second term in (4); it does not establish performance at untested sample ratios.

Archived-data results show no practical nonlinear gain. At $n = 2000$, cubic/PPI++ ratios are 1.0014 ([1.0013, 1.0015]) for galaxies, 0.9989 ([0.9988, 0.9991]) for forest, and 1.0015 ([1.0013, 1.0017]) for AlphaFold. Cubic coverage is 0.9488, 0.9474, and 0.9492, respectively; their Wilson intervals include .95. The small forest reduction is approximately one tenth of one percent; the other two widen. At $n = 500$ all three widen. These results concern the empirical distributions of archived labels and frozen predictions, with independent with-replacement draws. They support no claim of broader natural-population coverage or useful real-data shortening.

A narrow normal interval can fail badly. For rare Bernoulli outcomes with prevalence .01 and $n = 100$, cubic coverage is 0.6346 ([0.6212, 0.6478]), label-only Wald coverage is 0.6352 ([0.6218, 0.6484]), and exact Clopper–Pearson coverage is 0.9836 ([0.9797, 0.9868]). The apparent Wald

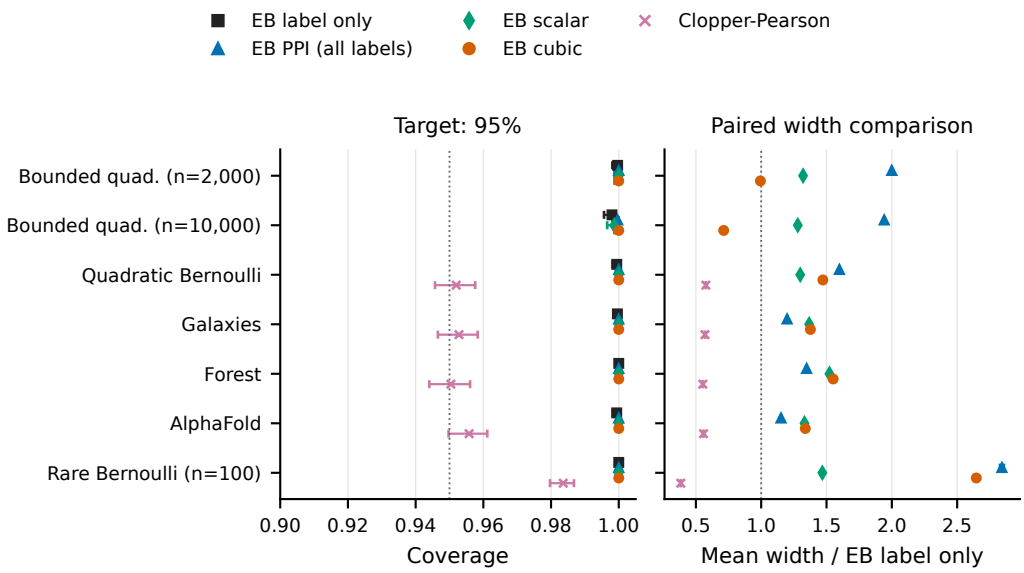


Figure 2: Finite-sample guarantees can coexist with gains or substantial conservatism. EB methods have bounded-data guarantees under independent iid sampling; trained controls reserve labels from the original budget, while label/fixed-PPI baselines use all n. CP is exact binary label-only and is absent for nonbinary bounded rows. All rows have $n = 2000$, $N = 20000$, except labeled $n = 10000$ bounded ($N = 100000$) and $n = 100$ rare-event ($N = 1000$) rows. Bounded rows use $R = 3000$; binary rows use $R = 5000$. Left: empirical coverage with 95% Wilson Monte Carlo intervals. Right: paired mean-width ratios to EB label-only with 95% delta-method Monte Carlo intervals. The designed bounded case gains; archived empirical populations do not. These Monte Carlo intervals are distinct from target-mean intervals and theorem guarantees. This compares the stated EB/CP family.

precision is inadmissible for the task's validity requirement. Zero-width intervals when all selected labels are zero remain in the summaries. The exploratory Holm-adjusted undercoverage diagnostic flags cubic intervals for galaxies and AlphaFold at $n = 100$, and rare Bernoulli at $n = 100, 500$. Failure to reject in other cells is not a coverage certificate.

Cross-fitting also leaves nuisance-fit instability: at $n = 100$, forest cubic widths have median .131636 and maximum 35.827096, while AlphaFold has median .101198 and maximum 8.748830. These large intervals are retained in means and uncertainty. Degree-12 and same-label plug-in diagnostics further expose fitting costs; they are not presented as automatically valid competitors.

Finite-sample guaranteed shortening is possible, but conservative. In the high-signal bounded-quadratic supplement, $n = 10000$, $N = 100000$, cubic EB/label-only EB mean-width ratio is 0.7128 ([0.7127, 0.7129]), a 28.7% reduction. Against all-label unit-PPI EB it is 0.3671 ([0.3671, 0.3672]), and against trained scalar EB it is 0.5568 ([0.5567, 0.5570]). Cubic EB mean width is 0.01343 (Monte Carlo SE 3.4×10^{-7}), and all 3,000 intervals cover (Wilson interval [0.9987, 1.0000]). The $\geq 95\%$ guarantee follows from boundedness and independent training, not this hit count. At $n = 2000$, the ratio to label-only EB is 0.9940 ([0.9937, 0.9943]): range penalties nearly erase the gain.

On all three empirical populations, cubic EB is wider than label-only EB, all-label unit-PPI EB, scalar EB, and exact Clopper–Pearson. In the nonlinear Bernoulli case it can beat unit-PPI EB while losing to label-only EB and Clopper–Pearson. Thus the finite-sample success is limited to a designed high-signal setting and the measured EB comparator family. These conservative intervals also remain substantially wider than their Wald counterparts; comparing across guarantee classes cannot remove that cost.

7 Discussion and limitations

The value of a frozen predictor depends on the information in its score, rather than its raw prediction accuracy alone. A scalar correction can leave nonlinear conditional-mean information unused. However, the extra fit consumes statistical precision, and at small label counts it can erase an oracle gain or invalidate a normal approximation. A no-signal score provides no mean-estimation advantage in the deterministic control class. An affine conditional mean leaves scalar tuning first-order optimal in that class.

Finite-sample validity imposes a different price. Known-range concentration and independent training provide a robust guarantee for bounded outcomes, but conservative range terms and label splitting can dominate variance savings. A finite-sample positive result in a designed high-signal regime does not establish improvement for all bounded populations, nor does empirical coverage validate an unproved Gaussian interval.

Our analysis is restricted to means and scalar frozen scores. It does not establish global optimality over all estimators or confidence sets, inference under distribution shift, conditional-on-the-observed-sample validity, robustness to adaptive basis selection, or generalization beyond the archived empirical populations. Synthetic mechanisms expose possibilities and failures under a known law; they cannot determine their prevalence in scientific practice. The practical lesson is to examine additional explainable variation, coefficient-estimation cost and coverage together before claiming precision gains.

Broader impacts. Using a frozen score more effectively can reduce the precision cost of expensive labeling. The main risk is overconfident scientific or policy inference when a short interval undercovers, or when an archived-data result is generalized beyond its sampling population. Reporting failures, separating guarantee classes and retaining reproducible coverage diagnostics are essential safeguards against that misuse; this work does not validate deployment in any particular scientific or policy setting.

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A Derivation of the variance comparisons

Write $Y = m(S) + \varepsilon$, with $\mathbb{E}[\varepsilon \mid S] = 0$. Orthogonality gives

$$\begin{aligned} nV(h) &= \mathbb{E}[\mathrm{Var}(Y \mid S)] + \mathrm{Var}(m - h) + r\,\mathrm{Var}(h) \\ &= \mathrm{Var}(Y) - 2\,\mathrm{Cov}(m, h) + (1 + r)\,\mathrm{Var}(h). \end{aligned}$$

Completing the square proves (3). Restricting h to a fixed linear span replaces centered m by its orthogonal projection into that span. Pythagoras then proves (6). The scalar case follows by differentiating the quadratic in a. When m is affine, centered m already belongs to the scalar span. When m is constant, every nonconstant control adds variance to the label-only estimator.

For the fitted-control result, assume a fixed number $K = 2$ of folds with positive limiting size fractions, fixed basis dimension, finite second moments of Y and $B(S)$, nonsingular population basis covariance, consistent fitted coefficients, $n/N \to r \in (0, \infty)$, and positive limiting oracle variance. Let γ be the population centered-regression coefficient, $b^* = \gamma/(1 + n/N)$, $w_k = n_k/n$, and $\widehat{b}_{-k}$ the coefficient fit outside fold k. The exact remainder relative to the deterministic-control estimator is

$$\widehat{\mu}_{\mathrm{cf}} - \widehat{\mu}_{B^\top b^*} = \sum_{k=1}^{2} w_k (\overline{B}_U - \overline{B}_{L_k})^\top (\widehat{b}_{-k} - b^*).$$

Each feature-mean difference is $O_p(n^{-1/2} + N^{-1/2})$ and each coefficient error is $o_p(1)$; therefore the finite sum is $o_p(n^{-1/2})$. Shared unlabeled observations across folds do not invalidate this product-order argument. Coefficient error itself need not be smaller than a sample-mean fluctuation. Consistency of coefficients and sample second moments makes the out-of-fold residual variance and averaged-control unlabeled variance converge to the oracle components. The independent-sample central limit theorem and Slutsky's theorem then give $(\widehat{\mu}_{\mathrm{cf}} - \mu)/\widehat{\mathrm{SE}} \Rightarrow \mathcal{N}(0, 1)$. This is an asymptotic result with stable dimension and consistent fitting; it supplies no finite-n guarantee for rare events or poorly conditioned fits.

B Finite-sample coverage proof

Condition on the training observations. The control is fixed and has its stated known range. The empirical Bernstein inequality applied in both directions to the evaluation residual mean fails with probability at most $\alpha/2$; the same holds for the independent unlabeled control mean. On their simultaneous event, their sum differs from μ by at most (8). The union bound gives conditional coverage at least $1 - \alpha$; taking expectation over training preserves it. Since $\mu \in [0, 1]$, intersecting the resulting interval with that known parameter space cannot remove the true mean. Independence from evaluation labels and correct support bounds are essential assumptions.

C Experimental and reproduction details

Synthetic populations. Independently draw all labeled and unlabeled scores uniformly on $[0, 1]$. Write $T = 2S - 1$; all noise terms below are independent of scores and of each other. The linear population has $Y = .2 + .8S + \epsilon$, $\epsilon \sim \mathcal{N}(0, .2^2)$, with mean .6. The quadratic population has $Y = T^2 + \epsilon$, $\epsilon \sim \mathcal{N}(0, .15^2)$, with mean $1/3$. The sigmoid population has $Y = (1 + \exp(-12(S - .5)))^{-1} + \epsilon$ with the same .15 Gaussian noise and mean .5. The no-signal population has $Y = .4 + \epsilon$, $\epsilon \sim \mathcal{N}(0, .2^2)$, with mean .4. Heavy-tailed quadratic data use $Y = T^2 + .15t_3$, with mean $1/3$ and finite noise variance $3(.15)^2$. Quadratic Bernoulli data use $Y \mid S \sim \mathrm{Bernoulli}(.05 + .9T^2)$ with mean .35. Rare binary data use $Y \sim \mathrm{Bernoulli}(.01)$ independently of S. The bounded supplement has $Y = .025 + .95T^2 + \epsilon$, $\epsilon \sim \mathrm{Uniform}[-.025, .025]$, with mean $.025 + .95/3$ and known support $[0, 1]$.

The bounded supplement was specified before inspecting the main results to test range penalties in a designed high-signal regime. The equal-n, N supplement followed inspection of initial real/bounded findings; it is a follow-up, not a preregistered original-grid experiment. No method or basis degree was selected by evaluation labels. The exact projection calculations, including their parameters, are generated by `code/theory.py`. The quadratic oracle/scalar ratio is

$$\sqrt{\frac{.15^2 + (4/45)r/(1 + r)}{.15^2 + 4/45}}, \qquad r = n/N.$$

The analytical values and their interpretation are presented separately from measured performance.

Empirical populations and data provenance. The PPI v0.2.3 example archives contain 16,743 galaxy records, 1,596 forest records and 10,802 AlphaFold records. Each provides binary archived labels and scores in $[0, 1]$. The empirical mean labels are .2592725318, .1516290727, and .1773745603, respectively. The labels are reference annotations for this benchmark, not asserted error-free physical truth. The original study descriptions and examples are in Angelopoulos et al. [2023]; we use neither all of Galaxy Zoo 2 nor a new scientific sampling frame. The archive's unlabeled sample is simulated by hiding outcomes of independent with-replacement draws, not by relabeling or expanding the labeled draw.

`provenance/manifest.json` gives pinned source URLs, byte sizes, SHA-256 values and versions. `code/download_data.py` checks existing files or retrieves the three exact package-declared Drive IDs. The reference package's code is MIT licensed; a separate license for its NPZ data is not established by the supplied documentation. We do not infer that the code's license covers the data; original data terms require checking before external redistribution. The official venue files retain their original provenance, and no separate template license is inferred.

Baseline implementation and fitting. The default scalar baseline reproduces `ppi-python==0.2.3`: covariance uses divisor n, pooled score variance uses divisor $n + N - 1$, the tuning denominator includes $1 + n/N$, and automatic tuning clips to $[0, 1]$. The unrestricted scalar removes only clipping. Normal standard-error components use divisor n or N, matching the reference. Twenty-four interval comparisons against the installed package agree with maximum endpoint difference 1.11×10^{-16} (`results/verification.json`). A Legendre coordinate $P_1(2S - 1) = 2S - 1$ therefore receives half the coefficient expressed on S; constants cancel.

The cubic uses P_1, P_2, P_3, centered least squares, two equal deterministic folds, and shrinkage $1/(1 + n/N)$. Least squares uses a Moore–Penrose solution with relative singular-value cutoff 10^{-10}. Intercepts cancel in the Wald correction. The degree-12 variants and same-label cubic are fixed diagnostics; there is no regularization or degree search. Numerically truncated or poorly conditioned fits can fail the coefficient-consistency condition; the asymptotic argument does not override that requirement. For EB, training uses $q = \max(10, \lfloor n/4 \rfloor)$ labels, requiring $n - q \geq 2$ and $N \geq 2$. The scalar coefficient is clipped to $[0, 1]$ before shrinkage; the cubic includes an intercept, clips predictions to $[0, 1]$ and then shrinks by $1/(1+(n-q)/N)$. These known ranges supply the guarantee. Label-only and all-label unit-control EB need no training split. All Wald intervals remain untruncated, including degenerate and extreme outcomes; EB and exact binary intervals use the known parameter range.

Seeds and uncertainty. The root seed is 20261102. A draw uses NumPy `SeedSequence([seed, scenario_ordinal, n, N/n, rep])`, where the fixed scenario order is linear, quadratic, sigmoid, no signal, heavy tail, quadratic Bernoulli, rare binary, galaxies, forest, AlphaFold, bounded quadratic. Repetition IDs start at zero. The internal output name `null` was renamed `no_signal` because standard CSV readers interpreted the former as missing data; its ordinal and measurements are unchanged. No incomplete or failed trial is removed from reported summaries.

For paired widths A_i, B_i and R repetitions, $\widehat{\rho} = \bar{A}/\bar{B}$ has Monte Carlo standard error

$$\widehat{\mathrm{SE}}(\widehat{\rho}) = \frac{\mathrm{sd}(A_i - \widehat{\rho} B_i)}{\sqrt{R}\,\bar{B}}.$$

Reported ratio intervals use $\widehat{\rho} \pm 1.96\widehat{\mathrm{SE}}$. Width-mean standard errors use the across-repetition sample standard deviation divided by $\sqrt{R}$. These quantify Monte Carlo estimation uncertainty, not a target-parameter confidence interval or a width-distribution percentile band. Coverage intervals are two-sided Wilson intervals for the integer hit counts. At coverage near .95, 5,000 repetitions give approximately .006 half-width, so smaller coverage differences cannot be resolved sharply. Exploratory exact-binomial tests and Holm adjustments are retained in `results/coverage_diagnostics.csv`.

Environment, commands and compute. The analyses used Python 3.11.2, NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.6 and PyArrow 26.0.0. Figure-generation versions are in `presentation/`

render_manifest.json; the manuscript uses TeX Live 2022, latexmk 4.79 and the unmodified official NeurIPS 2026 style. Run from the delivery root after installing code/requirements.txt, with OPENBLAS_NUM_THREADS=1 and OMP_NUM_THREADS=1. Exact reproduction commands are in analysis/README.md; the main commands are code/run_experiments.py with the synthetic/real groups, 5,000 repetitions and $n = 100, 500, 2000$, followed by exact/full-label EB baseline additions and code/summarize.py. Include all four result directories, including ratio1, when regenerating summaries and audits. The figure/table mapping and command are in presentation/README.md and presentation/captions.md. Build the paper with bash manuscript/build.sh; numerical prose is imported from the supplied summaries by manuscript/import_numbers.py, which records input hashes.

CPU-only jobs ran with the sandbox's two-CPU and 4096 MB memory ceilings on a shared host, without a GPU or paid API. Recorded elapsed times are 474.9 s for the main synthetic grid, 223.5 s for real data, 68.3 s for bounded data, and 19.3 s for the equal-n, N follow-up. Their sum is run elapsed time, not measured CPU consumption or total project time; shared-host load affects reruns. An approximate rerun budget is tens of minutes for the experiments and postprocessing, excluding downloads and environment setup. This is an estimate. Timing pilots and postprocessing are recorded separately in analysis/run_log.md.

All main experiments completed. Initial reference-package import failed because dependencies were missing, then succeeded after installation. A first finalization job was intentionally stopped after 20.668 s to replace slow repeated row assignments; completed baseline additions were retained idempotently. This was a postprocessing change, not exclusion or rerunning of unfavorable experiments. Extreme widths and undercoverage remain in all summaries; degree-12 diagnostics were not promoted to the primary method. No main experiment is represented by a pilot.

Source boundaries and LLM disclosure. Only the supplied starter papers, exact version-pinned reference package paths, the directly addressed unrelated empirical-Bernstein reference, and official format/ethics pages were consulted. Broad literature and citation-chain searches were avoided under the task's independence restriction. No prohibited source was sought or identified as encountered. The bounded source review cannot establish exhaustive priority or novelty; manuscript/source-consultation.md records the manuscript's sources. LLM-based research agents participated in study design, mathematical exposition, implementation, review and manuscript preparation. Reported empirical evidence comes from executed numerical code; no LLM-generated outcome labels or model predictions were used as experimental evidence.

D Analytical oracle limits

Table 1: Analytical leading-order width ratios within the score-control class. The oracle uses $h^* = \mathbb{E}[Y \mid S]/(1+n/N)$; the scalar uses its population-optimal unrestricted coefficient. Entries are square roots of exact variances from code/theory.py, rendered by presentation/render.py. They are theoretical predictions, not measured widths or finite-sample coverage guarantees.

Regime	n/N	Oracle / scalar	Oracle / label
Linear	0.1	1.000	0.693
Linear	1	1.000	0.845
Quadratic	0.1	0.524	0.524
Quadratic	1	0.775	0.775
No signal	0.1	1.000	1.000
No signal	1	1.000	1.000
Heavy-tailed noise	0.1	0.695	0.695
Heavy-tailed noise	1	0.846	0.846
Quadratic Bernoulli	0.1	0.844	0.844
Quadratic Bernoulli	1	0.917	0.917
Bounded quadratic	0.1	0.305	0.305
Bounded quadratic	1	0.708	0.708

E Complete evidence displays

All figures below target $\mu = \mathbb{E}[Y]$ with nominal 95% intervals. Coverage bars are 95% Wilson Monte Carlo intervals; width-ratio bars are paired delta-method 95% Monte Carlo intervals, not target-parameter intervals. Ratios divide paired mean widths, and values below one indicate narrowing. Every trial remains in the summaries. All figures and tables are generated by `presentation/render.py`; `presentation/captions.md` maps each output to its exact inputs and generator function, and `presentation/render_manifest.json` records their hashes.

Table 2: Large-label main-grid comparisons at $n = 2000, N = 20000, R = 5000$. Coverage includes 95% Wilson Monte Carlo intervals. Width ratio is cubic/PPI++ mean width, ± 1.96 paired Monte Carlo standard errors. Archive rows target empirical populations. Narrowing with undercoverage is not a valid improvement; compatibility with nominal coverage is not a proof. Absolute widths are retained in Appendix G.

Regime	PPI++ coverage	Cubic coverage	Width ratio
Linear	0.948 [0.941, 0.954]	0.948 [0.942, 0.954]	1.00119 ± 0.00004
Quadratic	0.949 [0.942, 0.954]	0.950 [0.944, 0.956]	0.52480 ± 0.00020
Sigmoid	0.945 [0.939, 0.951]	0.949 [0.942, 0.954]	0.85365 ± 0.00019
No signal	0.952 [0.946, 0.958]	0.952 [0.946, 0.958]	1.00145 ± 0.00005
Heavy-tailed noise	0.946 [0.939, 0.952]	0.950 [0.944, 0.956]	0.69245 ± 0.00098
Quadratic Bernoulli	0.948 [0.941, 0.953]	0.949 [0.943, 0.955]	0.84537 ± 0.00029
Rare Bernoulli	0.949 [0.942, 0.955]	0.943 [0.937, 0.949]	1.00140 ± 0.00005
Galaxies	0.949 [0.943, 0.955]	0.949 [0.942, 0.955]	1.00141 ± 0.00007
Forest	0.949 [0.943, 0.955]	0.947 [0.941, 0.953]	0.99894 ± 0.00014
AlphaFold	0.944 [0.937, 0.950]	0.949 [0.943, 0.955]	1.00150 ± 0.00015

Table 3: Coverage scope and additional conditions for implemented families. Independent iid draws from a common population underpin score-control comparisons; CP uses iid binary labels. Cross-fitting alone gives no finite-sample guarantee. EB uses known support ranges and the stated concentration allocation; trained controls must be independent of evaluation labels conditional on training.

Family	Additional conditions	Coverage scope
Label/PPI/scalar Wald	Controls h=0, S, or aS; finite second moments; consistent variance and applicable normal limit.	Asymptotic
Cross-fit fixed basis	Prespecified Legendre basis; fixed dimension; nonsingular covariance; consistent coefficients/variance; finite second moments.	Asymptotic
Bounded EB label/PPI	$Y, S \in [0, 1]$; known control range; independent common-population iid draws; stated error allocation.	Finite sample
Trained scalar/cubic EB	As above, with training independent of evaluation labels and bounded clipped control.	Finite sample
Clopper–Pearson	Binary iid labels from the target population. Uses no predictor.	Finite sample

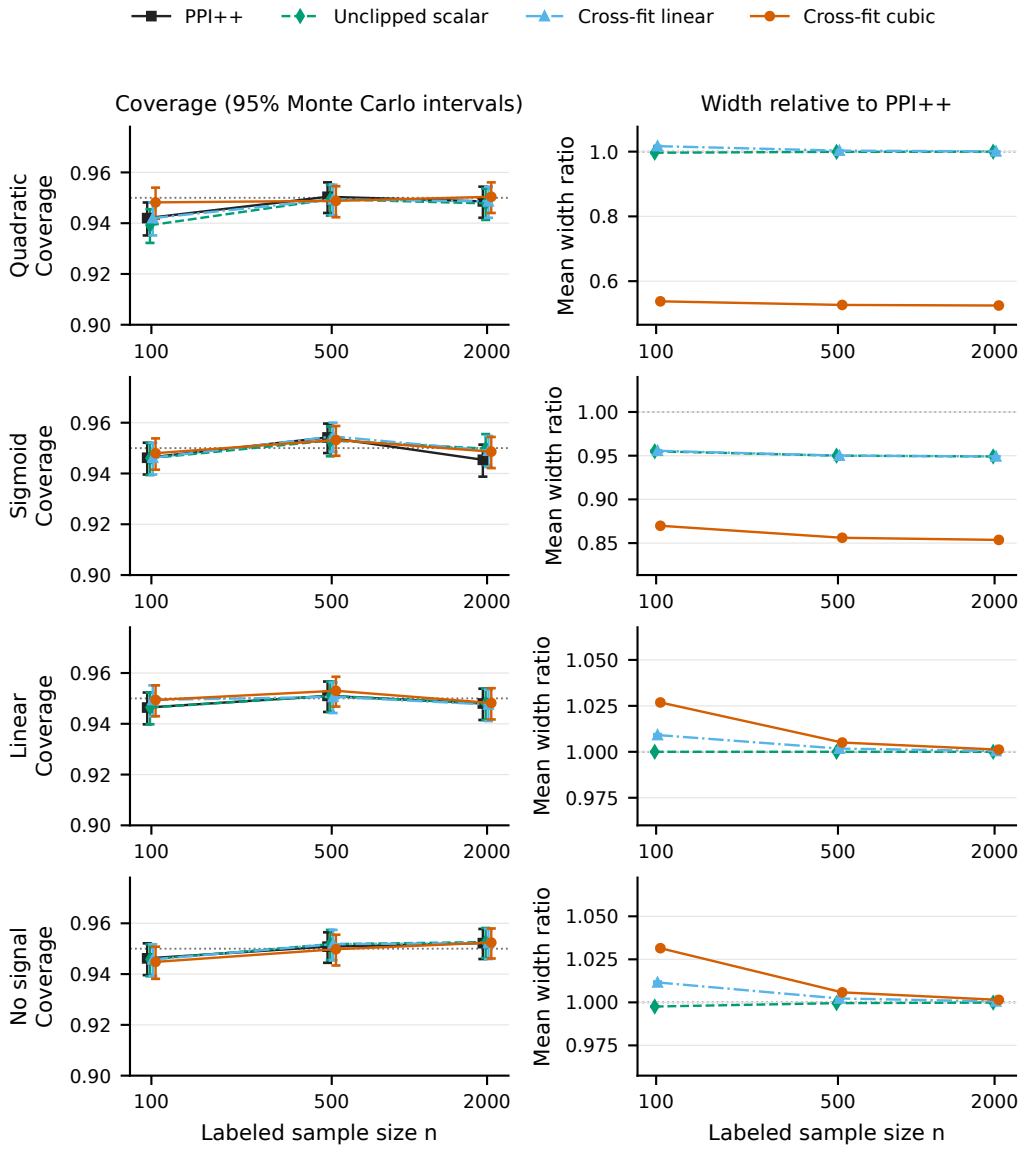


Figure 3: Dependence on label count in quadratic, sigmoid, affine and no-signal populations, with $N = 10n, R = 5000$ per setting. Coverage and paired mean-width ratios to PPI++ use the Monte Carlo intervals defined above. Unrestricted scalar and cross-fit linear controls distinguish nonlinear information from scalar clipping and fitting cost. Axes vary by row to retain all uncertainty endpoints and expose small costs alongside large gains; horizontal offsets separate overlapping methods. These are asymptotically justified Wald intervals.

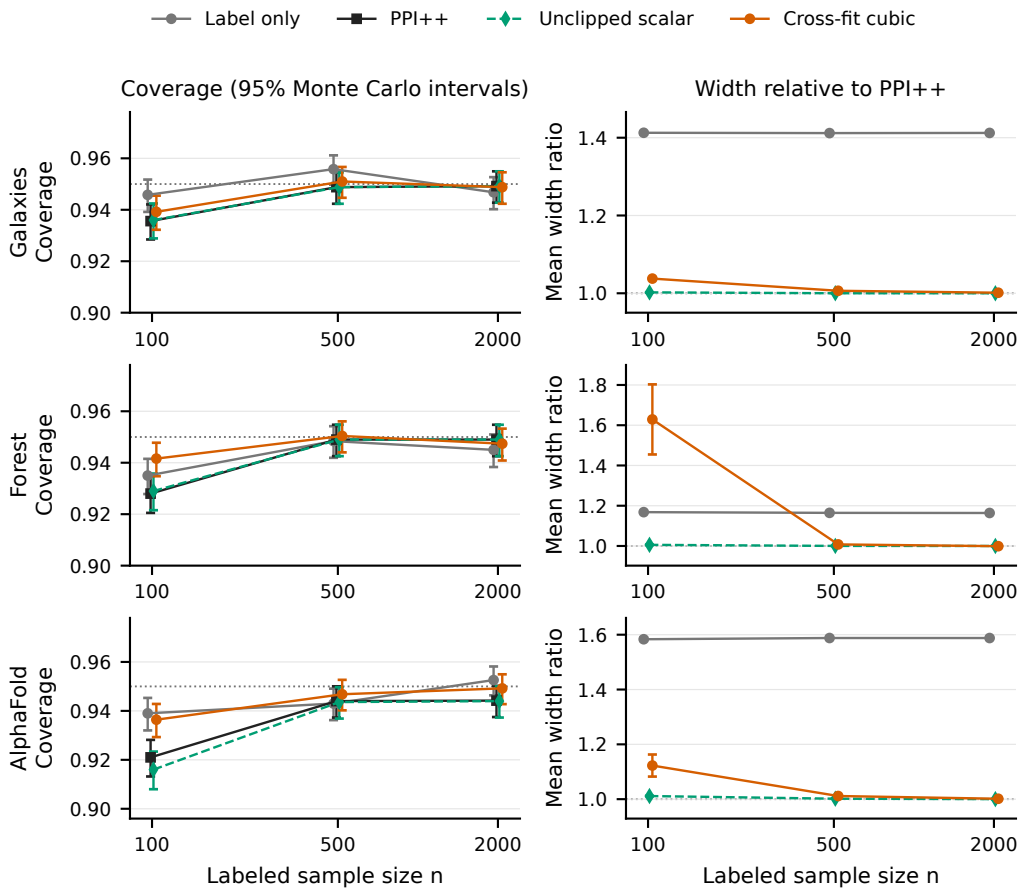


Figure 4: Archived-score results show small-label fitting costs and negligible large-label nonlinear gain. Independent with-replacement draws target each archive's empirical mean, with no broader population claim; $N = 10n, R = 5000$ per setting. Coverage uses Wilson 95% Monte Carlo intervals; mean-width ratios to PPI++ use paired delta-method 95% Monte Carlo intervals. Width means retain unstable large fits. These Wald intervals have asymptotic justification.

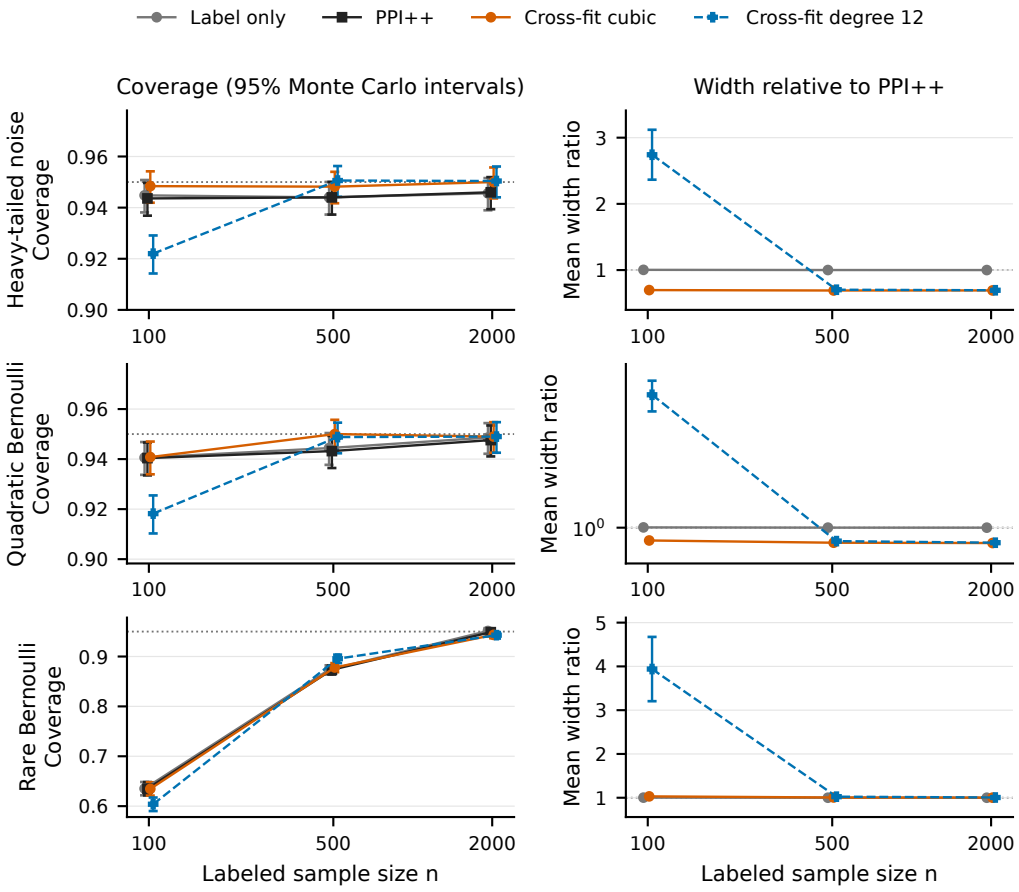


Figure 5: Heavy tails, nonlinear binary outcomes, rare events and a fixed degree-12 cross-fit diagnostic expose finite-sample limitations; $N = 10n, R = 5000$. Coverage uses Wilson 95% Monte Carlo intervals, and mean-width ratios to PPI++ use paired delta-method 95% Monte Carlo intervals. Cross-fitting does not ensure finite-sample validity or stable fitting, and short rare-event Wald intervals fail. The quadratic-Bernoulli width panel is logarithmic; other panels are linear. Axis ranges retain all uncertainty endpoints. Every trial remains in the means.

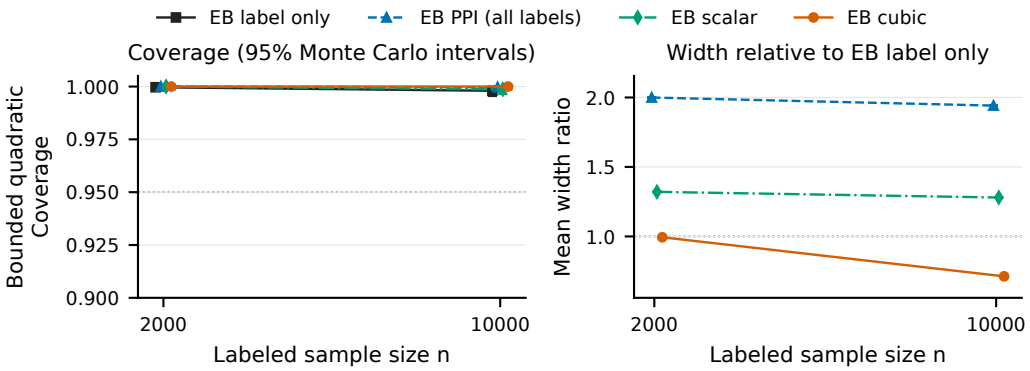


Figure 6: Sample-size dependence of the bounded high-signal quadratic EB comparison; $N = 10n, R = 3000$. All training consumes the original n labels. Coverage uses Wilson 95% Monte Carlo intervals, and mean-width ratios to EB label-only use paired delta-method 95% Monte Carlo intervals. At $n = 2000$ range penalties nearly erase the gain; at $n = 10000$ cubic EB is shorter while conservative. The $\geq 95\%$ guarantee follows from known-range concentration and independent training, independently of these hit counts.

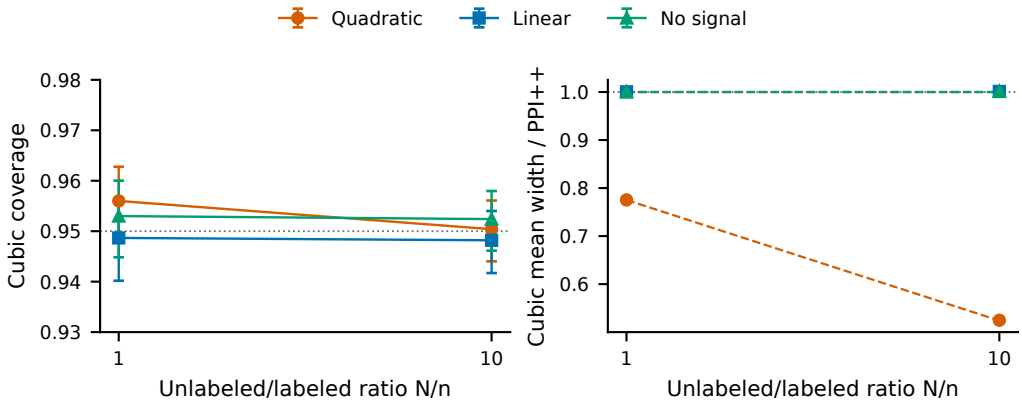


Figure 7: Finite unlabeled information limits the gain. At $n = 2000$, $N/n = 1$ uses $R = 3000$ (a declared follow-up) and $N/n = 10$ uses $R = 5000$ (main grid). Left: cubic coverage with Wilson 95% Monte Carlo intervals. Right: paired cubic/PPI++ mean-width ratios with delta-method 95% Monte Carlo intervals. Dashed lines are analytical oracle/unrestricted-scalar ratios, not empirical fits. Linear/no-signal oracle curves coincide at one. This is a class-restricted asymptotic comparison, not a universal interval-length bound. Methods share identical inputs within each setting.

F Finite-sample comparisons and fitting diagnostics

Table 4: Bounded-quadratic finite-sample comparison, $N = 10n, R = 3000$. Coverage has Wilson 95% Monte Carlo intervals. Width is the absolute mean in outcome units, with its Monte Carlo standard error in parentheses. Ratios to EB label-only report ± 1.96 paired Monte Carlo standard errors. Training uses the original budget; label-only and fixed-PPI use all n. Guarantees follow from assumptions, not observed coverage.

n	Method	Coverage	Mean (MCSE)	Width / EB label
2000	EB label only	1.000 [0.998, 1.000]	0.04779 (8.4×10^{-6})	1 (reference)
2000	EB PPI (full)	1.000 [0.999, 1.000]	0.09555 (1.8×10^{-5})	1.99949 $\pm$ 0.00070
2000	EB scalar	1.000 [0.999, 1.000]	0.06311 (2.1×10^{-5})	1.32075 $\pm$ 0.00076
2000	EB cubic	1.000 [0.999, 1.000]	0.0475 (1.7×10^{-6})	0.99400 $\pm$ 0.00032
10000	EB label only	0.998 [0.996, 0.999]	0.01884 (1.6×10^{-6})	1 (reference)
10000	EB PPI (full)	1.000 [0.998, 1.000]	0.03657 (3.6×10^{-6})	1.94153 $\pm$ 0.00035
10000	EB scalar	0.999 [0.997, 0.999]	0.02411 (3.3×10^{-6})	1.28012 $\pm$ 0.00027
10000	EB cubic	1.000 [0.999, 1.000]	0.01343 (3.4×10^{-7})	0.71279 $\pm$ 0.00011

Table 5: Exact binary CP and bounded EB comparisons, $N = 10n$, $R = 5000$. Five binary regimes at $n = 2000$ and rare events at $n = 100$ are retained. Coverage has Wilson 95% Monte Carlo intervals; absolute mean widths have Monte Carlo standard errors in parentheses. CP is exact for binary iid labels. Archive rows target empirical populations under independent with-replacement draws. Fitted EB gives no archive-data gain.

Regime	n	Method	Coverage	Mean (MCSE)
Quadratic Bernoulli	2000	CP	0.952 [0.946, 0.958]	0.04228 (4.2×10^{-6})
Quadratic Bernoulli	2000	EB label only	0.999 [0.998, 1.000]	0.07338 (6.4×10^{-6})
Quadratic Bernoulli	2000	EB PPI (full)	1.000 [0.999, 1.000]	0.1173 (1.5×10^{-5})
Quadratic Bernoulli	2000	EB scalar	1.000 [0.999, 1.000]	0.09531 (2.3×10^{-5})
Quadratic Bernoulli	2000	EB cubic	1.000 [0.999, 1.000]	0.108 (1.7×10^{-5})
Rare Bernoulli	100	CP	0.984 [0.980, 0.987]	0.0524 (2.0×10^{-4})
Rare Bernoulli	100	EB label only	1.000 [0.999, 1.000]	0.1362 (4.0×10^{-4})
Rare Bernoulli	100	EB PPI (full)	1.000 [0.999, 1.000]	0.3873 (4.8×10^{-4})
Rare Bernoulli	100	EB scalar	1.000 [0.999, 1.000]	0.2 (5.4×10^{-4})
Rare Bernoulli	100	EB cubic	1.000 [0.999, 1.000]	0.3605 (5.2×10^{-4})
Rare Bernoulli	2000	CP	0.960 [0.954, 0.965]	0.009243 (1.4×10^{-5})
Rare Bernoulli	2000	EB label only	1.000 [0.999, 1.000]	0.02158 (3.8×10^{-5})
Rare Bernoulli	2000	EB PPI (full)	1.000 [0.999, 1.000]	0.05068 (1.0×10^{-4})
Rare Bernoulli	2000	EB scalar	1.000 [0.999, 1.000]	0.02613 (5.1×10^{-5})
Rare Bernoulli	2000	EB cubic	1.000 [0.999, 1.000]	0.03409 (5.1×10^{-5})
Galaxies	2000	CP	0.953 [0.947, 0.958]	0.03888 (6.7×10^{-6})
Galaxies	2000	EB label only	1.000 [0.999, 1.000]	0.06825 (1.0×10^{-5})
Galaxies	2000	EB PPI (full)	1.000 [0.999, 1.000]	0.08179 (1.5×10^{-5})
Galaxies	2000	EB scalar	1.000 [0.999, 1.000]	0.09331 (2.2×10^{-5})
Galaxies	2000	EB cubic	1.000 [0.999, 1.000]	0.09395 (2.0×10^{-5})
Forest	2000	CP	0.950 [0.944, 0.956]	0.03192 (9.7×10^{-6})
Forest	2000	EB label only	1.000 [0.999, 1.000]	0.05772 (1.5×10^{-5})
Forest	2000	EB PPI (full)	1.000 [0.999, 1.000]	0.07779 (1.6×10^{-5})
Forest	2000	EB scalar	1.000 [0.999, 1.000]	0.08788 (3.1×10^{-5})
Forest	2000	EB cubic	1.000 [0.999, 1.000]	0.0895 (2.2×10^{-5})
AlphaFold	2000	CP	0.956 [0.950, 0.961]	0.03394 (8.8×10^{-6})
AlphaFold	2000	EB label only	0.999 [0.998, 1.000]	0.06077 (1.3×10^{-5})
AlphaFold	2000	EB PPI (full)	1.000 [0.999, 1.000]	0.06998 (1.7×10^{-5})
AlphaFold	2000	EB scalar	1.000 [0.999, 1.000]	0.08089 (2.2×10^{-5})
AlphaFold	2000	EB cubic	1.000 [0.999, 1.000]	0.08128 (2.3×10^{-5})

Table 6: Fitting diagnostics at $n = 100, N = 1000, R = 5000$. Width medians, empirical 99th percentiles and maxima are computed from saved draws. CF denotes cross-fitting. Extreme fits remain in every mean and uncertainty estimate; high-degree cross-fitting is a diagnostic, not an automatically valid alternative. Absolute widths are in outcome units and need not share a scale across populations. Archive rows target empirical populations.

Regime	Method	Median	99th percentile	Maximum
Quadratic	CF cubic	0.069	0.080	0.088
Quadratic	CF degree 12	0.090	2.717	33.471
Quadratic	Plug-in degree 12	0.066	0.085	0.894
No signal	CF cubic	0.080	0.096	0.117
No signal	CF degree 12	0.112	4.324	54.839
No signal	Plug-in degree 12	0.074	0.097	0.562
Galaxies	CF cubic	0.125	0.159	0.216
Galaxies	CF degree 12	0.183	232.088	258258.462
Galaxies	Plug-in degree 12	0.112	0.141	0.538
Forest	CF cubic	0.132	1.607	35.827
Forest	CF degree 12	17962.463	22647862.897	80256413.504
Forest	Plug-in degree 12	0.128	11520.016	3277521.013
AlphaFold	CF cubic	0.101	0.152	8.749
AlphaFold	CF degree 12	1.430	1544278.825	27793093.714
AlphaFold	Plug-in degree 12	0.086	0.345	25.303

G Every completed method/setting cell

The following tables retain all 405 cells, including all diagnostics and the equal-n, N follow-up. Full paired comparisons are in `tables/all_paired.csv`; `tables/all_cells.csv` is the lossless summary export. Both are generated from the analysis summaries by `presentation/render.py`. EB split and full-label PPI are distinct. Archive cells use independent with-replacement sampling of the empirical population; all other cells use the stated synthetic laws. Coverage intervals are two-sided 95% Wilson Monte Carlo intervals. Width is the mean; MCSE is its Monte Carlo standard error. All methods receive the same inputs per draw. Target coverage is 95%. CF denotes cross-fitting; CP denotes Clopper–Pearson. Coverage scope is given in the method definitions.

Table 7: Complete results: Linear.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.948 [0.941, 0.953]	0.11884	1.1×10^{-4}
100	1000	5000	PPI	0.952 [0.946, 0.957]	0.088596	7.4×10^{-5}
100	1000	5000	PPI++	0.946 [0.940, 0.952]	0.082367	7.5×10^{-5}
100	1000	5000	Scalar (unclip.)	0.946 [0.940, 0.952]	0.082368	7.5×10^{-5}
100	1000	5000	CF linear	0.949 [0.943, 0.955]	0.083119	7.9×10^{-5}
100	1000	5000	CF cubic	0.949 [0.943, 0.955]	0.084579	8.6×10^{-5}
100	1000	5000	Plug-in cubic	0.945 [0.938, 0.951]	0.081427	7.5×10^{-5}
100	1000	5000	CF degree 12	0.925 [0.918, 0.932]	0.34731	2.8×10^{-2}
100	1000	5000	Plug-in degree 12	0.888 [0.879, 0.896]	0.079282	1.1×10^{-4}
500	5000	5000	Label only	0.951 [0.944, 0.956]	0.053491	2.2×10^{-5}
500	5000	5000	PPI	0.951 [0.945, 0.957]	0.039775	1.5×10^{-5}
500	5000	5000	PPI++	0.951 [0.945, 0.957]	0.037051	1.5×10^{-5}
500	5000	5000	Scalar (unclip.)	0.951 [0.945, 0.957]	0.037051	1.5×10^{-5}
500	5000	5000	CF linear	0.951 [0.944, 0.956]	0.037116	1.5×10^{-5}
500	5000	5000	CF cubic	0.953 [0.947, 0.959]	0.037238	1.6×10^{-5}
500	5000	5000	Plug-in cubic	0.951 [0.945, 0.957]	0.036969	1.5×10^{-5}
500	5000	5000	CF degree 12	0.951 [0.944, 0.956]	0.037883	1.8×10^{-5}
500	5000	5000	Plug-in degree 12	0.944 [0.937, 0.950]	0.0367	1.5×10^{-5}
2000	2000	3000	Label only	0.953 [0.945, 0.960]	0.026765	6.9×10^{-6}
2000	2000	3000	PPI	0.954 [0.946, 0.961]	0.031179	4.7×10^{-6}
2000	2000	3000	PPI++	0.949 [0.941, 0.956]	0.022616	5.2×10^{-6}
2000	2000	3000	Scalar (unclip.)	0.949 [0.941, 0.956]	0.022616	5.2×10^{-6}
2000	2000	3000	CF linear	0.949 [0.940, 0.956]	0.022621	5.2×10^{-6}
2000	2000	3000	CF cubic	0.949 [0.940, 0.956]	0.022626	5.2×10^{-6}
2000	2000	3000	Plug-in cubic	0.949 [0.941, 0.956]	0.022614	5.2×10^{-6}
2000	20000	5000	Label only	0.953 [0.947, 0.958]	0.026774	5.3×10^{-6}
2000	20000	5000	PPI	0.948 [0.941, 0.953]	0.019929	3.7×10^{-6}
2000	20000	5000	PPI++	0.948 [0.941, 0.954]	0.018566	3.7×10^{-6}
2000	20000	5000	Scalar (unclip.)	0.948 [0.941, 0.954]	0.018566	3.7×10^{-6}
2000	20000	5000	CF linear	0.948 [0.941, 0.953]	0.018574	3.7×10^{-6}
2000	20000	5000	CF cubic	0.948 [0.942, 0.954]	0.018588	3.7×10^{-6}
2000	20000	5000	Plug-in cubic	0.947 [0.941, 0.953]	0.018556	3.7×10^{-6}
2000	20000	5000	CF degree 12	0.948 [0.941, 0.954]	0.018655	3.8×10^{-6}
2000	20000	5000	Plug-in degree 12	0.948 [0.941, 0.953]	0.018522	3.7×10^{-6}

Table 8: Complete results: Quadratic.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.946 [0.940, 0.952]	0.12983	1.1×10^{-4}
100	1000	5000	PPI	0.944 [0.937, 0.950]	0.17517	1.8×10^{-4}

Quadratic (continued)

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	PPI++	0.942 [0.935, 0.948]	0.12939	1.1×10^{-4}
100	1000	5000	Scalar (unclip.)	0.939 [0.932, 0.945]	0.12894	1.2×10^{-4}
100	1000	5000	CF linear	0.942 [0.935, 0.948]	0.13158	1.3×10^{-4}
100	1000	5000	CF cubic	0.948 [0.942, 0.954]	0.069574	6.0×10^{-5}
100	1000	5000	Plug-in cubic	0.945 [0.938, 0.951]	0.067449	5.3×10^{-5}
100	1000	5000	CF degree 12	0.919 [0.912, 0.927]	0.25494	1.5×10^{-2}
100	1000	5000	Plug-in degree 12	0.891 [0.882, 0.900]	0.066587	2.3×10^{-4}
500	5000	5000	Label only	0.952 [0.945, 0.957]	0.058422	2.3×10^{-5}
500	5000	5000	PPI	0.952 [0.946, 0.958]	0.078899	3.6×10^{-5}
500	5000	5000	PPI++	0.950 [0.944, 0.956]	0.058384	2.3×10^{-5}
500	5000	5000	Scalar (unclip.)	0.949 [0.943, 0.955]	0.058345	2.3×10^{-5}
500	5000	5000	CF linear	0.949 [0.943, 0.955]	0.05856	2.3×10^{-5}
500	5000	5000	CF cubic	0.949 [0.942, 0.955]	0.030742	1.1×10^{-5}
500	5000	5000	Plug-in cubic	0.948 [0.941, 0.954]	0.03056	1.0×10^{-5}
500	5000	5000	CF degree 12	0.948 [0.941, 0.953]	0.031173	1.2×10^{-5}
500	5000	5000	Plug-in degree 12	0.944 [0.938, 0.950]	0.030375	1.0×10^{-5}
2000	2000	3000	Label only	0.955 [0.947, 0.962]	0.029237	7.2×10^{-6}
2000	2000	3000	PPI	0.951 [0.943, 0.958]	0.046197	1.0×10^{-5}
2000	2000	3000	PPI++	0.955 [0.947, 0.962]	0.029234	7.3×10^{-6}
2000	2000	3000	Scalar (unclip.)	0.954 [0.946, 0.961]	0.029231	7.2×10^{-6}
2000	2000	3000	CF linear	0.955 [0.947, 0.962]	0.029245	7.3×10^{-6}
2000	2000	3000	CF cubic	0.956 [0.948, 0.963]	0.022668	4.5×10^{-6}
2000	2000	3000	Plug-in cubic	0.956 [0.948, 0.963]	0.022661	4.5×10^{-6}
2000	20000	5000	Label only	0.948 [0.942, 0.954]	0.029235	5.6×10^{-6}
2000	20000	5000	PPI	0.950 [0.944, 0.956]	0.039473	9.0×10^{-6}
2000	20000	5000	PPI++	0.949 [0.942, 0.954]	0.02923	5.6×10^{-6}
2000	20000	5000	Scalar (unclip.)	0.948 [0.941, 0.954]	0.029226	5.6×10^{-6}
2000	20000	5000	CF linear	0.949 [0.942, 0.954]	0.029254	5.7×10^{-6}
2000	20000	5000	CF cubic	0.950 [0.944, 0.956]	0.01534	2.7×10^{-6}
2000	20000	5000	Plug-in cubic	0.951 [0.944, 0.956]	0.015317	2.6×10^{-6}
2000	20000	5000	CF degree 12	0.950 [0.944, 0.956]	0.015385	2.7×10^{-6}
2000	20000	5000	Plug-in degree 12	0.949 [0.943, 0.955]	0.015294	2.6×10^{-6}

Table 9: Complete results: Sigmoid.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.946 [0.939, 0.952]	0.16955	9.8×10^{-5}
100	1000	5000	PPI	0.946 [0.940, 0.952]	0.090217	6.8×10^{-5}
100	1000	5000	PPI++	0.946 [0.940, 0.952]	0.090225	6.8×10^{-5}
100	1000	5000	Scalar (unclip.)	0.946 [0.939, 0.952]	0.086168	6.1×10^{-5}
100	1000	5000	CF linear	0.946 [0.940, 0.952]	0.086219	6.2×10^{-5}
100	1000	5000	CF cubic	0.948 [0.941, 0.954]	0.07848	5.8×10^{-5}
100	1000	5000	Plug-in cubic	0.945 [0.938, 0.951]	0.076328	4.9×10^{-5}
100	1000	5000	CF degree 12	0.920 [0.912, 0.927]	0.27418	1.5×10^{-2}
100	1000	5000	Plug-in degree 12	0.900 [0.892, 0.908]	0.074183	1.5×10^{-4}
500	5000	5000	Label only	0.947 [0.941, 0.953]	0.076215	2.0×10^{-5}
500	5000	5000	PPI	0.954 [0.948, 0.960]	0.040524	1.4×10^{-5}
500	5000	5000	PPI++	0.954 [0.948, 0.960]	0.040524	1.4×10^{-5}
500	5000	5000	Scalar (unclip.)	0.953 [0.947, 0.959]	0.0385	1.2×10^{-5}
500	5000	5000	CF linear	0.955 [0.948, 0.960]	0.038501	1.2×10^{-5}
500	5000	5000	CF cubic	0.953 [0.947, 0.959]	0.034692	1.0×10^{-5}
500	5000	5000	Plug-in cubic	0.952 [0.946, 0.958]	0.034516	9.9×10^{-6}

Sigmoid (continued)

n	N	R	Method	Coverage [95% MC]	Width	MCSE
500	5000	5000	CF degree 12	0.948 [0.941, 0.954]	0.03449	1.1×10^{-5}
500	5000	5000	Plug-in degree 12	0.947 [0.940, 0.952]	0.033778	9.5×10^{-6}
2000	20000	5000	Label only	0.949 [0.942, 0.954]	0.038149	4.8×10^{-6}
2000	20000	5000	PPI	0.945 [0.939, 0.951]	0.020284	3.5×10^{-6}
2000	20000	5000	PPI++	0.945 [0.939, 0.951]	0.020284	3.5×10^{-6}
2000	20000	5000	Scalar (unclip.)	0.950 [0.943, 0.956]	0.019252	3.1×10^{-6}
2000	20000	5000	CF linear	0.949 [0.943, 0.955]	0.019253	3.1×10^{-6}
2000	20000	5000	CF cubic	0.949 [0.942, 0.954]	0.017316	2.5×10^{-6}
2000	20000	5000	Plug-in cubic	0.949 [0.942, 0.955]	0.017294	2.5×10^{-6}
2000	20000	5000	CF degree 12	0.948 [0.941, 0.954]	0.017066	2.5×10^{-6}
2000	20000	5000	Plug-in degree 12	0.946 [0.939, 0.952]	0.016984	2.4×10^{-6}

Table 10: Complete results: No signal.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.948 [0.941, 0.954]	0.077812	7.9×10^{-5}
100	1000	5000	PPI	0.942 [0.935, 0.948]	0.14143	1.1×10^{-4}
100	1000	5000	PPI++	0.946 [0.940, 0.952]	0.077634	7.9×10^{-5}
100	1000	5000	Scalar (unclip.)	0.946 [0.939, 0.952]	0.077446	7.9×10^{-5}
100	1000	5000	CF linear	0.946 [0.939, 0.952]	0.078534	8.3×10^{-5}
100	1000	5000	CF cubic	0.945 [0.938, 0.951]	0.080079	9.1×10^{-5}
100	1000	5000	Plug-in cubic	0.940 [0.933, 0.947]	0.076721	7.9×10^{-5}
100	1000	5000	CF degree 12	0.920 [0.912, 0.927]	0.36701	2.4×10^{-2}
100	1000	5000	Plug-in degree 12	0.876 [0.867, 0.885]	0.074622	1.5×10^{-4}
500	5000	5000	Label only	0.952 [0.946, 0.958]	0.034998	1.5×10^{-5}
500	5000	5000	PPI	0.946 [0.939, 0.952]	0.063531	2.3×10^{-5}
500	5000	5000	PPI++	0.951 [0.944, 0.956]	0.034982	1.5×10^{-5}
500	5000	5000	Scalar (unclip.)	0.952 [0.946, 0.957]	0.034966	1.5×10^{-5}
500	5000	5000	CF linear	0.952 [0.946, 0.957]	0.035061	1.6×10^{-5}
500	5000	5000	CF cubic	0.950 [0.943, 0.956]	0.035185	1.6×10^{-5}
500	5000	5000	Plug-in cubic	0.950 [0.943, 0.955]	0.034902	1.5×10^{-5}
500	5000	5000	CF degree 12	0.949 [0.943, 0.955]	0.035867	1.9×10^{-5}
500	5000	5000	Plug-in degree 12	0.945 [0.938, 0.951]	0.034615	1.5×10^{-5}
2000	2000	3000	Label only	0.952 [0.943, 0.959]	0.017528	5.0×10^{-6}
2000	2000	3000	PPI	0.951 [0.943, 0.958]	0.039843	6.7×10^{-6}
2000	2000	3000	PPI++	0.952 [0.944, 0.959]	0.017527	5.0×10^{-6}
2000	2000	3000	Scalar (unclip.)	0.952 [0.943, 0.959]	0.017526	5.0×10^{-6}
2000	2000	3000	CF linear	0.952 [0.944, 0.959]	0.017531	5.0×10^{-6}
2000	2000	3000	CF cubic	0.953 [0.945, 0.960]	0.017537	5.0×10^{-6}
2000	2000	3000	Plug-in cubic	0.952 [0.944, 0.959]	0.017521	5.0×10^{-6}
2000	20000	5000	Label only	0.952 [0.946, 0.958]	0.017527	3.9×10^{-6}
2000	20000	5000	PPI	0.953 [0.946, 0.958]	0.031805	5.7×10^{-6}
2000	20000	5000	PPI++	0.952 [0.946, 0.958]	0.017525	3.9×10^{-6}
2000	20000	5000	Scalar (unclip.)	0.953 [0.946, 0.958]	0.017523	3.9×10^{-6}
2000	20000	5000	CF linear	0.952 [0.946, 0.958]	0.017535	3.9×10^{-6}
2000	20000	5000	CF cubic	0.952 [0.946, 0.958]	0.01755	4.0×10^{-6}
2000	20000	5000	Plug-in cubic	0.952 [0.946, 0.958]	0.017515	3.9×10^{-6}
2000	20000	5000	CF degree 12	0.950 [0.943, 0.955]	0.01762	4.0×10^{-6}
2000	20000	5000	Plug-in degree 12	0.949 [0.943, 0.955]	0.017479	3.9×10^{-6}

Table 11: Complete results: Heavy-tailed noise.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.945 [0.938, 0.951]	0.1523	3.4×10^{-4}
100	1000	5000	PPI	0.944 [0.938, 0.950]	0.19278	3.5×10^{-4}
100	1000	5000	PPI++	0.944 [0.937, 0.950]	0.1518	3.4×10^{-4}
100	1000	5000	Scalar (unclip.)	0.942 [0.935, 0.948]	0.15131	3.4×10^{-4}
100	1000	5000	CF linear	0.945 [0.938, 0.951]	0.15407	3.5×10^{-4}
100	1000	5000	CF cubic	0.948 [0.942, 0.954]	0.1059	4.1×10^{-4}
100	1000	5000	Plug-in cubic	0.940 [0.933, 0.947]	0.10206	3.9×10^{-4}
100	1000	5000	CF degree 12	0.922 [0.914, 0.929]	0.41628	2.9×10^{-2}
100	1000	5000	Plug-in degree 12	0.884 [0.875, 0.893]	0.099977	4.5×10^{-4}
500	5000	5000	Label only	0.944 [0.937, 0.950]	0.068817	8.1×10^{-5}
500	5000	5000	PPI	0.949 [0.942, 0.954]	0.086921	7.5×10^{-5}
500	5000	5000	PPI++	0.944 [0.937, 0.950]	0.068775	8.1×10^{-5}
500	5000	5000	Scalar (unclip.)	0.943 [0.936, 0.949]	0.068731	8.1×10^{-5}
500	5000	5000	CF linear	0.943 [0.937, 0.949]	0.068977	8.1×10^{-5}
500	5000	5000	CF cubic	0.948 [0.942, 0.954]	0.047553	9.9×10^{-5}
500	5000	5000	Plug-in cubic	0.950 [0.944, 0.956]	0.047211	9.8×10^{-5}
500	5000	5000	CF degree 12	0.951 [0.944, 0.956]	0.048357	1.0×10^{-4}
500	5000	5000	Plug-in degree 12	0.945 [0.939, 0.951]	0.046866	9.7×10^{-5}
2000	20000	5000	Label only	0.946 [0.939, 0.952]	0.034549	2.9×10^{-5}
2000	20000	5000	PPI	0.951 [0.944, 0.956]	0.043576	2.6×10^{-5}
2000	20000	5000	PPI++	0.946 [0.939, 0.952]	0.034544	2.9×10^{-5}
2000	20000	5000	Scalar (unclip.)	0.946 [0.940, 0.952]	0.034538	2.9×10^{-5}
2000	20000	5000	CF linear	0.946 [0.940, 0.952]	0.034571	2.9×10^{-5}
2000	20000	5000	CF cubic	0.950 [0.944, 0.956]	0.02392	3.6×10^{-5}
2000	20000	5000	Plug-in cubic	0.950 [0.944, 0.956]	0.023877	3.6×10^{-5}
2000	20000	5000	CF degree 12	0.950 [0.944, 0.956]	0.024007	3.6×10^{-5}
2000	20000	5000	Plug-in degree 12	0.951 [0.944, 0.956]	0.023834	3.6×10^{-5}

Table 12: Complete results: Quadratic Bernoulli.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.941 [0.934, 0.947]	0.18582	8.7×10^{-5}
100	1000	5000	PPI	0.949 [0.943, 0.955]	0.22017	1.8×10^{-4}
100	1000	5000	PPI++	0.940 [0.933, 0.947]	0.18526	8.9×10^{-5}
100	1000	5000	Scalar (unclip.)	0.939 [0.932, 0.945]	0.18466	9.1×10^{-5}
100	1000	5000	CF linear	0.942 [0.936, 0.949]	0.18799	1.1×10^{-4}
100	1000	5000	CF cubic	0.941 [0.934, 0.947]	0.16106	1.6×10^{-4}
100	1000	5000	Plug-in cubic	0.933 [0.926, 0.939]	0.15453	1.4×10^{-4}
100	1000	5000	CF degree 12	0.918 [0.910, 0.925]	0.79299	6.7×10^{-2}
100	1000	5000	Plug-in degree 12	0.879 [0.869, 0.887]	0.15134	4.1×10^{-4}
100	1000	5000	EB label only	1.000 [0.999, 1.000]	0.48859	1.4×10^{-4}
100	1000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.80728	8.8×10^{-4}
100	1000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.91115	9.4×10^{-4}
100	1000	5000	EB scalar	1.000 [0.999, 1.000]	0.69725	1.0×10^{-3}
100	1000	5000	EB cubic	1.000 [0.999, 1.000]	0.84925	7.7×10^{-4}
100	1000	5000	CP	0.957 [0.951, 0.962]	0.19344	8.5×10^{-5}
500	5000	5000	Label only	0.944 [0.938, 0.950]	0.08351	1.7×10^{-5}
500	5000	5000	PPI	0.947 [0.941, 0.953]	0.098886	3.6×10^{-5}
500	5000	5000	PPI++	0.943 [0.936, 0.949]	0.083462	1.7×10^{-5}
500	5000	5000	Scalar (unclip.)	0.942 [0.935, 0.948]	0.083413	1.7×10^{-5}
500	5000	5000	CF linear	0.943 [0.936, 0.949]	0.083706	1.8×10^{-5}

Quadratic Bernoulli (continued)

n	N	R	Method	Coverage [95% MC]	Width	MCSE
500	5000	5000	CF cubic	0.950 [0.944, 0.956]	0.070802	2.8×10^{-5}
500	5000	5000	Plug-in cubic	0.948 [0.941, 0.954]	0.070285	2.7×10^{-5}
500	5000	5000	CF degree 12	0.949 [0.942, 0.955]	0.072049	3.6×10^{-5}
500	5000	5000	Plug-in degree 12	0.944 [0.937, 0.950]	0.069755	2.7×10^{-5}
500	5000	5000	EB label only	1.000 [0.999, 1.000]	0.16724	2.6×10^{-5}
500	5000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.28446	6.0×10^{-5}
500	5000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.34072	8.1×10^{-5}
500	5000	5000	EB scalar	1.000 [0.999, 1.000]	0.22666	1.4×10^{-4}
500	5000	5000	EB cubic	1.000 [0.999, 1.000]	0.28043	6.9×10^{-5}
500	5000	5000	CP	0.947 [0.941, 0.953]	0.085308	1.7×10^{-5}
2000	20000	5000	Label only	0.949 [0.942, 0.954]	0.0418	4.2×10^{-6}
2000	20000	5000	PPI	0.944 [0.938, 0.950]	0.049494	9.0×10^{-6}
2000	20000	5000	PPI++	0.948 [0.941, 0.953]	0.041794	4.2×10^{-6}
2000	20000	5000	Scalar (unclip.)	0.948 [0.941, 0.954]	0.041788	4.2×10^{-6}
2000	20000	5000	CF linear	0.948 [0.941, 0.953]	0.041824	4.3×10^{-6}
2000	20000	5000	CF cubic	0.949 [0.943, 0.955]	0.035331	6.8×10^{-6}
2000	20000	5000	Plug-in cubic	0.949 [0.943, 0.955]	0.035265	6.8×10^{-6}
2000	20000	5000	CF degree 12	0.949 [0.943, 0.955]	0.035459	7.0×10^{-6}
2000	20000	5000	Plug-in degree 12	0.949 [0.942, 0.955]	0.035199	6.8×10^{-6}
2000	20000	5000	EB label only	0.999 [0.998, 1.000]	0.073382	6.4×10^{-6}
2000	20000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.1173	1.5×10^{-5}
2000	20000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.1375	2.0×10^{-5}
2000	20000	5000	EB scalar	1.000 [0.999, 1.000]	0.09531	2.3×10^{-5}
2000	20000	5000	EB cubic	1.000 [0.999, 1.000]	0.10802	1.7×10^{-5}
2000	20000	5000	CP	0.952 [0.946, 0.958]	0.042275	4.2×10^{-6}

Table 13: Complete results: Rare Bernoulli.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.635 [0.622, 0.648]	0.030206	3.5×10^{-4}
100	1000	5000	PPI	0.948 [0.941, 0.954]	0.12404	1.2×10^{-4}
100	1000	5000	PPI++	0.635 [0.622, 0.648]	0.030133	3.5×10^{-4}
100	1000	5000	Scalar (unclip.)	0.635 [0.622, 0.648]	0.030064	3.4×10^{-4}
100	1000	5000	CF linear	0.635 [0.621, 0.648]	0.030475	3.5×10^{-4}
100	1000	5000	CF cubic	0.635 [0.621, 0.648]	0.031068	3.6×10^{-4}
100	1000	5000	Plug-in cubic	0.635 [0.621, 0.648]	0.029781	3.4×10^{-4}
100	1000	5000	CF degree 12	0.604 [0.590, 0.617]	0.11871	1.1×10^{-2}
100	1000	5000	Plug-in degree 12	0.607 [0.593, 0.620]	0.028912	3.4×10^{-4}
100	1000	5000	EB label only	1.000 [0.999, 1.000]	0.13624	4.0×10^{-4}
100	1000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.38726	4.8×10^{-4}
100	1000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.48274	5.6×10^{-4}
100	1000	5000	EB scalar	1.000 [0.999, 1.000]	0.20003	5.4×10^{-4}
100	1000	5000	EB cubic	1.000 [0.999, 1.000]	0.36055	5.2×10^{-4}
100	1000	5000	CP	0.984 [0.980, 0.987]	0.052395	2.0×10^{-4}
500	5000	5000	Label only	0.873 [0.864, 0.882]	0.016955	5.8×10^{-5}
500	5000	5000	PPI	0.951 [0.945, 0.957]	0.055796	2.3×10^{-5}
500	5000	5000	PPI++	0.873 [0.864, 0.882]	0.016947	5.8×10^{-5}
500	5000	5000	Scalar (unclip.)	0.874 [0.864, 0.883]	0.016939	5.8×10^{-5}
500	5000	5000	CF linear	0.874 [0.865, 0.883]	0.016984	5.8×10^{-5}
500	5000	5000	CF cubic	0.878 [0.869, 0.887]	0.017044	5.8×10^{-5}
500	5000	5000	Plug-in cubic	0.875 [0.865, 0.884]	0.016909	5.8×10^{-5}
500	5000	5000	CF degree 12	0.896 [0.887, 0.904]	0.017359	6.0×10^{-5}

Rare Bernoulli (continued)

n	N	R	Method	Coverage [95% MC]	Width	MCSE
500	5000	5000	Plug-in degree 12	0.881 [0.872, 0.889]	0.016771	5.7×10^{-5}
500	5000	5000	EB label only	1.000 [0.999, 1.000]	0.043327	1.1×10^{-4}
500	5000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.11614	2.0×10^{-4}
500	5000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.13882	2.3×10^{-4}
500	5000	5000	EB scalar	1.000 [0.999, 1.000]	0.057879	1.4×10^{-4}
500	5000	5000	EB cubic	1.000 [0.999, 1.000]	0.089673	1.4×10^{-4}
500	5000	5000	CP	0.981 [0.977, 0.984]	0.019518	5.4×10^{-5}
2000	20000	5000	Label only	0.951 [0.945, 0.957]	0.00868	1.4×10^{-5}
2000	20000	5000	PPI	0.951 [0.945, 0.957]	0.027926	6.0×10^{-6}
2000	20000	5000	PPI++	0.949 [0.942, 0.955]	0.008679	1.4×10^{-5}
2000	20000	5000	Scalar (unclip.)	0.947 [0.940, 0.953]	0.008678	1.4×10^{-5}
2000	20000	5000	CF linear	0.944 [0.937, 0.950]	0.0086838	1.4×10^{-5}
2000	20000	5000	CF cubic	0.943 [0.937, 0.949]	0.0086911	1.4×10^{-5}
2000	20000	5000	Plug-in cubic	0.944 [0.937, 0.950]	0.0086741	1.4×10^{-5}
2000	20000	5000	CF degree 12	0.943 [0.936, 0.949]	0.0087253	1.4×10^{-5}
2000	20000	5000	Plug-in degree 12	0.940 [0.933, 0.946]	0.0086561	1.4×10^{-5}
2000	20000	5000	EB label only	1.000 [0.999, 1.000]	0.021582	3.8×10^{-5}
2000	20000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.050683	1.0×10^{-4}
2000	20000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.057912	1.2×10^{-4}
2000	20000	5000	EB scalar	1.000 [0.999, 1.000]	0.026127	5.1×10^{-5}
2000	20000	5000	EB cubic	1.000 [0.999, 1.000]	0.034095	5.1×10^{-5}
2000	20000	5000	CP	0.960 [0.954, 0.965]	0.0092434	1.4×10^{-5}

Table 14: Complete results: Galaxies.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.946 [0.939, 0.952]	0.17086	1.4×10^{-4}
100	1000	5000	PPI	0.943 [0.936, 0.949]	0.12171	1.7×10^{-4}
100	1000	5000	PPI++	0.936 [0.928, 0.942]	0.12095	1.6×10^{-4}
100	1000	5000	Scalar (unclip.)	0.936 [0.929, 0.942]	0.12123	1.6×10^{-4}
100	1000	5000	CF linear	0.940 [0.933, 0.947]	0.12152	1.7×10^{-4}
100	1000	5000	CF cubic	0.939 [0.932, 0.945]	0.12551	1.9×10^{-4}
100	1000	5000	Plug-in cubic	0.930 [0.923, 0.937]	0.11878	1.7×10^{-4}
100	1000	5000	CF degree 12	0.943 [0.937, 0.949]	76.385	5.2×10^{1}
100	1000	5000	Plug-in degree 12	0.869 [0.860, 0.878]	0.11245	2.1×10^{-4}
100	1000	5000	EB label only	1.000 [0.999, 1.000]	0.46139	3.4×10^{-4}
100	1000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.63758	5.2×10^{-4}
100	1000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.7326	6.2×10^{-4}
100	1000	5000	EB scalar	1.000 [0.999, 1.000]	0.70494	6.9×10^{-4}
100	1000	5000	EB cubic	1.000 [0.999, 1.000]	0.72688	6.6×10^{-4}
100	1000	5000	CP	0.961 [0.956, 0.966]	0.17887	1.3×10^{-4}
500	5000	5000	Label only	0.956 [0.950, 0.961]	0.076717	2.7×10^{-5}
500	5000	5000	PPI	0.949 [0.943, 0.955]	0.054714	3.5×10^{-5}
500	5000	5000	PPI++	0.949 [0.942, 0.955]	0.054337	3.3×10^{-5}
500	5000	5000	Scalar (unclip.)	0.949 [0.942, 0.955]	0.054342	3.3×10^{-5}
500	5000	5000	CF linear	0.950 [0.944, 0.956]	0.054367	3.3×10^{-5}
500	5000	5000	CF cubic	0.951 [0.945, 0.957]	0.054685	3.4×10^{-5}
500	5000	5000	Plug-in cubic	0.949 [0.943, 0.955]	0.05412	3.3×10^{-5}
500	5000	5000	CF degree 12	0.951 [0.945, 0.957]	0.05598	3.8×10^{-5}
500	5000	5000	Plug-in degree 12	0.941 [0.934, 0.947]	0.053471	3.3×10^{-5}
500	5000	5000	EB label only	1.000 [0.999, 1.000]	0.15697	4.0×10^{-5}
500	5000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.21342	6.1×10^{-5}

Galaxies (continued)

n	N	R	Method	Coverage [95% MC]	Width	MCSE
500	5000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.25802	8.0×10^{-5}
500	5000	5000	EB scalar	1.000 [0.999, 1.000]	0.24809	9.5×10^{-5}
500	5000	5000	EB cubic	1.000 [0.999, 1.000]	0.25228	8.1×10^{-5}
500	5000	5000	CP	0.961 [0.955, 0.966]	0.07855	2.6×10^{-5}
2000	20000	5000	Label only	0.947 [0.940, 0.953]	0.038401	6.7×10^{-6}
2000	20000	5000	PPI	0.950 [0.944, 0.956]	0.027395	8.6×10^{-6}
2000	20000	5000	PPI++	0.949 [0.943, 0.955]	0.027191	8.2×10^{-6}
2000	20000	5000	Scalar (unclip.)	0.949 [0.943, 0.955]	0.027191	8.2×10^{-6}
2000	20000	5000	CF linear	0.948 [0.941, 0.954]	0.027193	8.2×10^{-6}
2000	20000	5000	CF cubic	0.949 [0.942, 0.955]	0.027229	8.3×10^{-6}
2000	20000	5000	Plug-in cubic	0.948 [0.942, 0.954]	0.027162	8.2×10^{-6}
2000	20000	5000	CF degree 12	0.949 [0.943, 0.955]	0.027361	8.5×10^{-6}
2000	20000	5000	Plug-in degree 12	0.948 [0.941, 0.953]	0.02707	8.2×10^{-6}
2000	20000	5000	EB label only	1.000 [0.999, 1.000]	0.068247	1.0×10^{-5}
2000	20000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.081791	1.5×10^{-5}
2000	20000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.096184	2.0×10^{-5}
2000	20000	5000	EB scalar	1.000 [0.999, 1.000]	0.093312	2.2×10^{-5}
2000	20000	5000	EB cubic	1.000 [0.999, 1.000]	0.093953	2.0×10^{-5}
2000	20000	5000	CP	0.953 [0.947, 0.958]	0.038881	6.7×10^{-6}

Table 15: Complete results: Forest.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.935 [0.928, 0.942]	0.13929	2.0×10^{-4}
100	1000	5000	PPI	0.939 [0.932, 0.945]	0.12046	2.0×10^{-4}
100	1000	5000	PPI++	0.928 [0.921, 0.935]	0.11924	1.9×10^{-4}
100	1000	5000	Scalar (unclip.)	0.929 [0.922, 0.936]	0.11991	2.0×10^{-4}
100	1000	5000	CF linear	0.943 [0.936, 0.949]	0.12295	2.2×10^{-4}
100	1000	5000	CF cubic	0.942 [0.935, 0.948]	0.19423	1.1×10^{-2}
100	1000	5000	Plug-in cubic	0.918 [0.910, 0.925]	0.11677	2.1×10^{-4}
100	1000	5000	CF degree 12	0.929 [0.922, 0.936]	1.2×10^{6}	6.1×10^{4}
100	1000	5000	Plug-in degree 12	0.543 [0.529, 0.557]	3719.2	1.2×10^{3}
100	1000	5000	EB label only	1.000 [0.999, 1.000]	0.36053	6.4×10^{-4}
100	1000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.52031	5.6×10^{-4}
100	1000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.61646	6.7×10^{-4}
100	1000	5000	EB scalar	1.000 [0.999, 1.000]	0.56665	1.0×10^{-3}
100	1000	5000	EB cubic	1.000 [0.999, 1.000]	0.61223	7.3×10^{-4}
100	1000	5000	CP	0.949 [0.943, 0.955]	0.14821	1.9×10^{-4}
500	5000	5000	Label only	0.948 [0.942, 0.954]	0.062758	3.8×10^{-5}
500	5000	5000	PPI	0.950 [0.943, 0.955]	0.054353	3.9×10^{-5}
500	5000	5000	PPI++	0.949 [0.943, 0.955]	0.05389	3.8×10^{-5}
500	5000	5000	Scalar (unclip.)	0.949 [0.943, 0.955]	0.053908	3.8×10^{-5}
500	5000	5000	CF linear	0.951 [0.945, 0.957]	0.054072	3.9×10^{-5}
500	5000	5000	CF cubic	0.950 [0.944, 0.956]	0.054309	4.0×10^{-5}
500	5000	5000	Plug-in cubic	0.949 [0.943, 0.955]	0.05345	3.8×10^{-5}
500	5000	5000	CF degree 12	0.949 [0.943, 0.955]	0.073629	4.5×10^{-3}
500	5000	5000	Plug-in degree 12	0.939 [0.932, 0.946]	0.052609	3.9×10^{-5}
500	5000	5000	EB label only	1.000 [0.999, 1.000]	0.13587	5.8×10^{-5}
500	5000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.20531	6.6×10^{-5}
500	5000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.25022	9.2×10^{-5}
500	5000	5000	EB scalar	1.000 [0.999, 1.000]	0.23422	1.6×10^{-4}
500	5000	5000	EB cubic	1.000 [0.999, 1.000]	0.24353	9.5×10^{-5}

Forest (continued)

n	N	R	Method	Coverage [95% MC]	Width	MCSE
500	5000	5000	CP	0.964 [0.958, 0.969]	0.064672	3.8×10^{-5}
2000	20000	5000	Label only	0.945 [0.938, 0.951]	0.031433	9.7×10^{-6}
2000	20000	5000	PPI	0.948 [0.942, 0.954]	0.027247	9.8×10^{-6}
2000	20000	5000	PPI++	0.949 [0.943, 0.955]	0.027006	9.6×10^{-6}
2000	20000	5000	Scalar (unclip.)	0.949 [0.943, 0.955]	0.027006	9.6×10^{-6}
2000	20000	5000	CF linear	0.949 [0.943, 0.955]	0.027027	9.6×10^{-6}
2000	20000	5000	CF cubic	0.947 [0.941, 0.953]	0.026977	9.6×10^{-6}
2000	20000	5000	Plug-in cubic	0.949 [0.942, 0.954]	0.026872	9.5×10^{-6}
2000	20000	5000	CF degree 12	0.949 [0.942, 0.955]	0.027041	9.8×10^{-6}
2000	20000	5000	Plug-in degree 12	0.945 [0.938, 0.951]	0.026685	9.5×10^{-6}
2000	20000	5000	EB label only	1.000 [0.999, 1.000]	0.057719	1.5×10^{-5}
2000	20000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.077794	1.6×10^{-5}
2000	20000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.092386	2.2×10^{-5}
2000	20000	5000	EB scalar	1.000 [0.999, 1.000]	0.087875	3.1×10^{-5}
2000	20000	5000	EB cubic	1.000 [0.999, 1.000]	0.089503	2.2×10^{-5}
2000	20000	5000	CP	0.950 [0.944, 0.956]	0.031923	9.7×10^{-6}

Table 16: Complete results: AlphaFold.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
100	1000	5000	Label only	0.939 [0.932, 0.945]	0.14851	1.8×10^{-4}
100	1000	5000	PPI	0.939 [0.932, 0.945]	0.093581	1.9×10^{-4}
100	1000	5000	PPI++	0.921 [0.913, 0.928]	0.09379	1.9×10^{-4}
100	1000	5000	Scalar (unclip.)	0.916 [0.908, 0.923]	0.094855	1.9×10^{-4}
100	1000	5000	CF linear	0.940 [0.933, 0.946]	0.094298	1.9×10^{-4}
100	1000	5000	CF cubic	0.936 [0.929, 0.943]	0.10529	1.9×10^{-3}
100	1000	5000	Plug-in cubic	0.916 [0.908, 0.924]	0.090717	2.0×10^{-4}
100	1000	5000	CF degree 12	0.957 [0.951, 0.962]	73641	1.2×10^{4}
100	1000	5000	Plug-in degree 12	0.751 [0.739, 0.763]	0.10579	5.8×10^{-3}
100	1000	5000	EB label only	1.000 [0.999, 1.000]	0.3921	6.4×10^{-4}
100	1000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.52856	4.6×10^{-4}
100	1000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.62011	5.5×10^{-4}
100	1000	5000	EB scalar	1.000 [0.999, 1.000]	0.5978	6.3×10^{-4}
100	1000	5000	EB cubic	1.000 [0.999, 1.000]	0.61735	6.2×10^{-4}
100	1000	5000	CP	0.952 [0.946, 0.958]	0.15715	1.8×10^{-4}
500	5000	5000	Label only	0.943 [0.936, 0.949]	0.066866	3.6×10^{-5}
500	5000	5000	PPI	0.947 [0.941, 0.953]	0.042161	3.8×10^{-5}
500	5000	5000	PPI++	0.944 [0.937, 0.950]	0.04211	3.7×10^{-5}
500	5000	5000	Scalar (unclip.)	0.944 [0.937, 0.950]	0.04218	3.7×10^{-5}
500	5000	5000	CF linear	0.947 [0.941, 0.953]	0.042096	3.7×10^{-5}
500	5000	5000	CF cubic	0.947 [0.940, 0.953]	0.042593	3.9×10^{-5}
500	5000	5000	Plug-in cubic	0.943 [0.937, 0.949]	0.041762	3.8×10^{-5}
500	5000	5000	CF degree 12	0.943 [0.936, 0.949]	0.044297	4.7×10^{-5}
500	5000	5000	Plug-in degree 12	0.929 [0.921, 0.936]	0.040977	3.8×10^{-5}
500	5000	5000	EB label only	1.000 [0.999, 1.000]	0.14208	5.5×10^{-5}
500	5000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.18976	6.7×10^{-5}
500	5000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.23121	8.8×10^{-5}
500	5000	5000	EB scalar	1.000 [0.999, 1.000]	0.22428	9.0×10^{-5}
500	5000	5000	EB cubic	1.000 [0.999, 1.000]	0.22715	9.3×10^{-5}
500	5000	5000	CP	0.957 [0.951, 0.962]	0.068755	3.6×10^{-5}
2000	20000	5000	Label only	0.953 [0.946, 0.958]	0.03345	8.9×10^{-6}
2000	20000	5000	PPI	0.946 [0.939, 0.952]	0.021116	9.4×10^{-6}

AlphaFold (continued)

n	N	R	Method	Coverage [95% MC]	Width	MCSE
2000	20000	5000	PPI++	0.944 [0.937, 0.950]	0.021066	9.3×10^{-6}
2000	20000	5000	Scalar (unclip.)	0.944 [0.937, 0.950]	0.021069	9.3×10^{-6}
2000	20000	5000	CF linear	0.949 [0.942, 0.954]	0.021059	9.3×10^{-6}
2000	20000	5000	CF cubic	0.949 [0.943, 0.955]	0.021098	9.6×10^{-6}
2000	20000	5000	Plug-in cubic	0.947 [0.940, 0.953]	0.020996	9.5×10^{-6}
2000	20000	5000	CF degree 12	0.946 [0.940, 0.952]	0.02122	9.9×10^{-6}
2000	20000	5000	Plug-in degree 12	0.945 [0.938, 0.951]	0.020859	9.5×10^{-6}
2000	20000	5000	EB label only	0.999 [0.998, 1.000]	0.060767	1.3×10^{-5}
2000	20000	5000	EB PPI (full)	1.000 [0.999, 1.000]	0.069982	1.7×10^{-5}
2000	20000	5000	EB PPI (split)	1.000 [0.999, 1.000]	0.082753	2.2×10^{-5}
2000	20000	5000	EB scalar	1.000 [0.999, 1.000]	0.080887	2.2×10^{-5}
2000	20000	5000	EB cubic	1.000 [0.999, 1.000]	0.081285	2.3×10^{-5}
2000	20000	5000	CP	0.956 [0.950, 0.961]	0.033937	8.8×10^{-6}

Table 17: Complete results: Bounded quadratic.

n	N	R	Method	Coverage [95% MC]	Width	MCSE
2000	20000	3000	Label only	0.943 [0.934, 0.951]	0.024859	5.5×10^{-6}
2000	20000	3000	PPI	0.950 [0.942, 0.958]	0.036356	1.1×10^{-5}
2000	20000	3000	PPI++	0.941 [0.932, 0.949]	0.024854	5.5×10^{-6}
2000	20000	3000	Scalar (unclip.)	0.941 [0.932, 0.949]	0.02485	5.5×10^{-6}
2000	20000	3000	CF linear	0.941 [0.932, 0.949]	0.024876	5.6×10^{-6}
2000	20000	3000	CF cubic	0.950 [0.942, 0.958]	0.0075918	5.1×10^{-7}
2000	20000	3000	Plug-in cubic	0.951 [0.942, 0.958]	0.0075914	5.1×10^{-7}
2000	20000	3000	EB label only	1.000 [0.998, 1.000]	0.047787	8.4×10^{-6}
2000	20000	3000	EB PPI (full)	1.000 [0.999, 1.000]	0.09555	1.8×10^{-5}
2000	20000	3000	EB PPI (split)	1.000 [0.999, 1.000]	0.11237	2.4×10^{-5}
2000	20000	3000	EB scalar	1.000 [0.999, 1.000]	0.063115	2.1×10^{-5}
2000	20000	3000	EB cubic	1.000 [0.999, 1.000]	0.0475	1.7×10^{-6}
10000	100000	3000	Label only	0.949 [0.941, 0.957]	0.011117	1.1×10^{-6}
10000	100000	3000	PPI	0.945 [0.937, 0.953]	0.016258	2.1×10^{-6}
10000	100000	3000	PPI++	0.950 [0.941, 0.957]	0.011116	1.1×10^{-6}
10000	100000	3000	Scalar (unclip.)	0.949 [0.940, 0.956]	0.011116	1.1×10^{-6}
10000	100000	3000	CF linear	0.950 [0.941, 0.957]	0.011118	1.1×10^{-6}
10000	100000	3000	CF cubic	0.952 [0.944, 0.959]	0.0033952	1.0×10^{-7}
10000	100000	3000	Plug-in cubic	0.952 [0.944, 0.959]	0.0033952	1.0×10^{-7}
10000	100000	3000	EB label only	0.998 [0.996, 0.999]	0.018837	1.6×10^{-6}
10000	100000	3000	EB PPI (full)	1.000 [0.998, 1.000]	0.036573	3.6×10^{-6}
10000	100000	3000	EB PPI (split)	1.000 [0.999, 1.000]	0.042142	4.7×10^{-6}
10000	100000	3000	EB scalar	0.999 [0.997, 0.999]	0.024114	3.3×10^{-6}
10000	100000	3000	EB cubic	1.000 [0.999, 1.000]	0.013427	3.4×10^{-7}

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